The 300mm Analog Fortress

A step-by-step narrative masterclass and audio companion to Texas Instruments: why 30-year 300mm analog wafer fabs in Sherman and Richardson beat Intel-style logic capex, lessons from Analog Devices, Infineon, and Fastenal, the $7.32 billion free-cash-flow inflection, and our disciplined support entry plan.

Company & Market: Texas Instruments Incorporated (NASDAQ: TXN) — Deep Investment & Business Monograph (US)
Research Session Cutoff: 2026-11-06 • **Generated:** 07 Nov 2026, 10:21 AM PST
Core Verdict: WAIT_FOR_SUPPORT (WAIT_FOR_30W_SMA_RECLAIM_OR_200DMA_SUPPORT)
Authored & Narrated by: Senior Institutional Equity Research Desk • Masterclass Companion Audio-Reader Edition (16,466 words • ~106 min listen)

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SESSION PRICE

$277.36

VERDICT

WAIT_FOR_SUPPORT

STAGE POSTURE

STAGE_3_PULLBACK_TO_50DMA

BUY / SUPPORT ZONE

$262.28–279.09

BASE / BULL TARGET

$325.00 / $368.00

PREFACE & CORE THESIS

Texas Instruments leads global analog ($16.82B TTM sales, 81.0% of revenue) and embedded ($3.03B TTM, 14.6%) chips across 80,000 parts for 100,000 customers. In Q3 2026 (reported 21 October 2026), revenue rose 28% YoY to $6.10B as Industrial jumped 40% and Data Center surged 95%, lifting gross margin to 64.0% and TTM free cash flow to $7.32B ($8.40B with CHIPS Act cash). At $278.59 ($254.57B market cap, 36.15x TTM P/E, 23.90x forward P/E), shares sit 16.6% below their $334.03 high. Price tests the 50-day SMA ($279.09), sitting 1.80% below the 30-week SMA ($283.70) and 6.22% above the rising 200-day SMA ($262.28) with RSI at 43.50. Our call is WAIT_FOR_SUPPORT in $262.28–$279.09 (requiring a weekly close above $283.70 or a 200-day SMA bounce, stop $255.80) toward a $325.00 base target and $368.00 bull target.

PROLOGUE

The Price, The Setup & Our Independent Verdict on Texas Instruments

> *"The best measure of a company's performance over time is the sustainable growth of free cash flow per share. That is because free cash flow is the only currency that can be reinvested in the moat or returned to the owners."*
>
> — RICHARD K. TEMPLETON, FORMER CHAIRMAN & CEO OF TEXAS INSTRUMENTS

If you are reading these pages or listening to them narrated aloud, imagine that the closing bells in New York have fallen silent, the Wall Street talking heads have finished arguing over next week's GPU shipments. Also, you and I are sitting across a desk with a warm cup of coffee looking at the quiet giant that powers every single piece of electronics on earth. On the surface of the financial news cycle, artificial intelligence is treated as a story of a few glamorous digital processors that cost thirty thousand dollars apiece. Yet if you walk onto an institutional semiconductor desk and open the schematic of an AI data center rack, an automated factory robot, or an 800-volt electric vehicle, you discover a completely different reality. None of those digital brains can wake up, draw a single amp of regulated electrical current, sense heat, or move a mechanical joint without hundreds of tiny, fifty-cent to three-dollar analog and embedded chips made by Texas Instruments Incorporated, trading on the NASDAQ under the ticker symbol TXN.

On November 6, 2026, Texas Instruments stands at one of the most fascinating crossroads you and I will ever study in public equity markets: the exact intersection of a powerful multi-year fundamental inflection and a sharp, intermediate technical pullback. Just sixteen days ago, on October 21, 2026, Chairman and Chief Executive Officer Haviv Ilan, newly appointed Chief Financial Officer Julie Knecht, and Head of Investor Relations Mike Beckman stepped up to the microphone to deliver the company's third-quarter 2026 financial results. For nearly three years prior to that evening, skeptics on Wall Street had doubted Texas Instruments. They wondered whether the painful post-pandemic inventory correction of 2023 and 2024 would drag on forever, and whether management's bold decision to pour thirty billion dollars into internal 300-millimeter wafer fabrication plants in Texas and Utah would crush returns on capital.

NARRATOR'S REFLECTION • THE DIFFERENCE BETWEEN A STORY STOCK AND A CASH MACHINE

"Wall Street falls in love with companies that promise to invent tomorrow's technology at any cost. Institutional compounders fall in love with companies that already own the physical toll bridges of today's economy and have just finished paying for the concrete."

When you and I study Texas Instruments at $278.59, we are looking at a rare moment in the capital cycle. For four years, TXN depressed its own reported free cash flow to build 300mm wafer fabs while competitors froze. Now the concrete is poured, the cleanrooms are standing, CapEx is falling by billions of dollars a year, and every incremental dollar of industrial and AI data center revenue is dropping through at nearly ninety cents of gross profit.

The October 21 earnings print answered those doubts not with vague promises, but with audited cash and operating leverage. Third-quarter revenue surged 28 percent year-over-year and 11 percent sequentially to reach 6.10 billion dollars. That single quarter lifted trailing twelve-month revenue to 20,766 million dollars, officially eclipsing the company's previous pandemic-era peak of 20,028 million dollars set back in fiscal 2022. Even more revealing was where that growth came from. Industrial. That is Texas Instruments' highest-margin flagship end market, roared ahead by more than 40 percent year-over-year and roughly 10 percent quarter-on-quarter, setting a brand-new all-time quarterly revenue record for the first time in four years. Meanwhile, the Data Center vertical skyrocketed 95 percent year-over-year and 20 percent sequentially to reach roughly 13 percent of total corporate revenue, or an annualized run-rate near 2.5 billion dollars.

Because Texas Instruments fabricates the overwhelming majority of its incremental silicon inside its own 300-millimeter cleanrooms in Richardson, Texas, Lehi, Utah, and Sherman, Texas, that top-line rebound flowed down the income statement with breathtaking force. As CFO Julie Knecht told listeners on the call, third-quarter gross profit reached 3.90 billion dollars, or 64.0 percent of revenue, jumping 280 basis points in a single quarter and delivering nearly 90 percent incremental gross margin on sequential revenue gains. Operating profit climbed 73 percent year-over-year to 2.90 billion dollars. That represents a 47.0 percent operating margin. Meanwhile, diluted earnings per share reached 2.62 dollars even after absorbing a one-cent discrete tax headwind.

And then came the number that truly matters to long-term owners like you and me: the free cash flow harvest. As major 300-millimeter cleanroom shells reached completion, capital expenditures stepped down to 463 million dollars in the third quarter and 2,578 million dollars over the trailing twelve months, down from 5,071 million dollars in fiscal 2023. Paired with a record 9,896 million dollars of trailing operating cash flow, standard trailing free cash flow leaped 251.8 percent year-over-year to 7,318 million dollars ($7.99 per share), or 8.40 billion dollars including 1.60 billion dollars of CHIPS Act tax credits and grants.

GLOBAL PARALLEL (USA & EUROPE) • INTEL VS. ANALOG DEVICES VS. TEXAS INSTRUMENTS — THREE PATHS IN SILICON

"Not all semiconductor capital expenditure is created equal: a leading-edge logic fab is a melting ice cube. By contrast, a 300mm analog fab is a thirty-year hydroelectric dam."

Generalist investors often flinch when they hear the words 'semiconductor fab capex' because they remember Intel (NASDAQ: INTC). Intel spent over $100 billion chasing 7nm and 18A digital logic nodes with $380 million ASML EUV lithography machines that become obsolete every three to four years, destroying shareholder wealth when designs slipped. At the opposite extreme, US peer Analog Devices (NASDAQ: ADI) chose a 'fab-lite' model after buying Linear Technology and Maxim Integrated, outsourcing most wafer production to Taiwan's TSMC (NYSE: TSM). ADI protected its cash flow during the 2023–2024 trough, but sacrificed the 40% unpackaged die-cost advantage that TI's captive 300mm fabs command. Only Germany's Infineon Technologies (ETR: IFX), with its 300mm Dresden Smart Power Fab, mirrors TI's insight: in analog and power semiconductors built on 45nm to 130nm nodes, the lithography tools run profitably for 25 to 30 years.

- Intel (NASDAQ: INTC): 3-to-4 year tool obsolescence on leading-edge logic. Negative free cash flow and severe ROIC destruction.
- Analog Devices (NASDAQ: ADI): Hybrid fab-lite model outsourcing to TSMC. 10.25% ROIC and higher wafer COGS during the 2026–2030 upcycle.
- Texas Instruments (NASDAQ: TXN): Captive 300mm analog fabs (Richardson, Lehi, Sherman) with 25–30 year equipment life, 40% lower die cost, and industry-leading 22.04% ROIC.

The Valuation Optical Illusion and Why the Stock Pulled Back to $278.59

If the fundamental inflection in the third quarter of 2026 was so decisive, why is Texas Instruments trading at 278 dollars and 59 cents on November 6, 2026, down 16.60 percent from its 52-week peak of 334 dollars and 3 cents and down 9.44 percent over the past five trading days? Between its 52-week low of 152 dollars and 73 cents and its autumn peak of 334 dollars and 3 cents, Texas Instruments had already surged 118.7 percent. Over the past week, a sharp rotation out of semiconductor equities dragged Microchip Technology down 25.08 percent and Analog Devices down 14.05 percent in five sessions, pulling Texas Instruments back to test its intermediate support lines.

At the same time, a glance at a standard financial terminal creates an optical valuation illusion. On trailing twelve-month GAAP figures through September 30, 2026, Texas Instruments trades at 36.15 times trailing earnings of 7.72 dollars per share, 34.64 times trailing standard free cash flow, and 23.97 times EV-to-EBITDA, with a trailing GAAP dividend payout ratio of 100.97 percent and 13.65 billion dollars of gross debt.

Yet trailing twelve-month accounting numbers blend earlier recovery quarters with the newly established earnings run-rate. In the third quarter of 2026 alone, Texas Instruments earned 2.62 dollars per share ($10.48 annualized). Management also guided the fourth quarter of 2026, a winter quarter that historically drops five to eight percent sequentially, to 5.80 billion to 6.30 billion dollars in revenue and 2.45 to 2.79 dollars in EPS ($2.62 midpoint). Against consensus fiscal 2027 earnings of roughly 11.65 dollars per share (+28.43 percent forward EPS growth), Texas Instruments trades at 23.90 times forward earnings and a PEG ratio of just 0.68.

> **MENTAL MODEL • THE REARVIEW MIRROR TRAP IN CYCLICAL INFLECTIONS**
>
> At the exact moment an industrial or semiconductor company emerges from a capital cycle trough, its trailing twelve-month P/E ratio always looks artificially high because the denominator contains depressed trough quarters while the numerator reflects forward recovery. Evaluating TXN on $7.72 of trailing EPS when Q3 and Q4 2026 are already running at $10.48 annualized. And heading toward $11.65 in FY2027. Is like driving a highway Ferrari while staring exclusively into the rearview mirror.

Walking Through Our Four-Lens Opportunity Matrix and No-FOMO Verdict

How should you and I translate this clash between accelerating fundamentals and a 16.6 percent technical pullback into a capital allocation decision? In the companion web dossier, we lay out a Four-Lens Institutional Opportunity Matrix across every investor horizon.

Lens One is Short-Term Trading and Momentum over one to twelve weeks. Here, our verdict is strictly Wait for Support. At 278 dollars and 59 cents, Texas Instruments sits 1.79 percent below its 30-week simple moving average ($283.68) and 1.25 percent below its 10-week exponential moving average ($282.13), resting on its 50-day simple moving average ($279.08) and Classic S1 pivot ($277.54). Meanwhile, only 17 percent of semiconductor stocks trade above their 50-day exponential moving average. Short-term traders should wait either for a Friday weekly close reclaiming 283 dollars and 70 cents on volume above 7.43 million shares, or a clean bounce off the rising 200-day moving average at 262 dollars and 28 cents, with a stop at 255 dollars and 80 cents and a swing target at 318 dollars and 60 cents.

Lens Two is Medium-Term Cyclical and Margin Inflection over three to eighteen months. Here, our posture is Accumulate on Support inside 262 dollars and 28 cents to 279 dollars and 9 cents. Three catalysts converge for fiscal 2027: an above-seasonal fourth-quarter revenue guide ($5.80B–$6.30B), the breakout in Industrial (+40 percent YoY) and Data Center (+95 percent YoY to ~$2.5B annualized), and active 2027 contract pricing negotiations. Accumulating on support positions us for our twelve-month base target of 325 dollars per share (28x FY27E EPS) and our bull target of 368 dollars.

Lens Three is Long-Term Compounding over three to ten years, and Lens Four is our Red-Team Stress Audit. Over a multi-year horizon, Texas Instruments is a Fortress Analog Compounder worthy of a 7.0 percent net-worth allocation under our Tier-2 Core Alpha risk budget, with room to compound up to our 22 percent runaway-winner ceiling. With over 80,000 catalog parts, more than 70 percent direct sales, and a 40 percent die-cost advantage on 300-millimeter wafers, TI is entering an 8 billion to 11 billion dollar annual free cash flow harvest while our Red-Team audit sets a 20 percent probability bear floor at 225 dollars.

THE EXECUTION PLAYBOOK • OUR EXACT BUY BAND, STOP-LOSS & TARGET COORDINATES

Discipline means writing down your exact price coordinates while the market is closed so emotion never hijacks your execution during market hours.

- Current Session Close (6 Nov 2026): $278.59 ($254.57B Market Cap | 23.90x Forward P/E | 0.68x PEG).
- Accumulation / Support Zone: $262.28 (rising 200-day SMA & 40-week EMA) to $279.09 (50-day SMA).
- Strength Trigger: Friday weekly close >= $283.70 reclaiming the 30-week SMA on volume > 7.43M shares.
- Hard Invalidation Stop: $255.80 (-8.18% risk from $278.59. 0.5x ATR14 below the rising 200-day SMA).
- 12-Month Targets & Sizing: $325.00 Base Target (+18.8% total return w/ $6.08 dividend), $368.00 Bull Target, $225.00 Bear Floor. 7.0% Net Worth TIER_2 CORE_ALPHA target weight.

CHAPTER I

From Oilfield Seismography & Jack Kilby's Integrated Circuit to the 300mm Analog Fortress

> *"What we did in the summer of 1958 was show that you did not have to wire separate components together by hand. That an entire circuit could live on a single monolithic piece of silicon."*
>
> — JACK S. KILBY, TEXAS INSTRUMENTS ENGINEER & 2000 NOBEL LAUREATE IN PHYSICS

To understand why Texas Instruments commands a quarter-trillion-dollar valuation and a 64 percent gross margin in late 2026, you and I have to travel back ninety-six years to the oilfields of Texas. Great corporate moats do not appear by accident. They are forged through decades of institutional memory and a willingness to walk away from glamorous, low-return businesses in favor of enduring manufacturing advantages. In December 1930, right in the depths of the Great Depression, J. Clarence 'Doc' Karcher and Eugene McDermott founded Geophysical Service Incorporated, or GSI, in Dallas, Texas. Their breakthrough was reflection seismography: detonating small charges in the earth and using sensitive analog acoustic instruments to measure how sound waves bounced off deep underground rock layers to locate oil traps.

When World War Two broke out, the United States Navy and Army Signal Corps realized that the exact same analog signal-processing expertise used to listen for underground oil could be used to detect German U-boats beneath the Atlantic and track aircraft on radar. Led by four legendary builders. Eugene McDermott, Cecil Green, J. Erik Jonsson, and Patrick Haggerty. The company expanded rapidly into precision military electronics. Recognizing after the war that solid-state electronics would eclipse oilfield contracting, the partners reorganized the company, renamed it Texas Instruments Incorporated in 1951, and listed its shares on the New York Stock Exchange on October 1, 1953.

Within five years of that NYSE listing, engineers inside Texas Instruments' Dallas laboratories achieved two breakthroughs that gave birth to the modern information age. First, in May 1954, TI research director Gordon Teal grew the world's first commercial silicon transistor. Until that day, early transistors were made of germanium. This stopped

working whenever a radio or aircraft cockpit grew warm. Teal's silicon transistor proved that semiconductors could operate reliably across extreme temperatures, powering both the famous Regency TR-1 pocket radio and United States missile guidance systems.

Then came the hot Texas summer of 1958. A newly hired engineer named Jack Kilby had not yet accrued enough vacation time to leave Dallas during the company's annual July shutdown. Working alone in a quiet laboratory, Kilby confronted what engineers called the 'tyranny of numbers'. The fact that complex computers and rockets required tens of thousands of discrete transistors, resistors, and capacitors soldered together by hand, where a single faulty solder joint doomed the entire machine. On September 12, 1958, standing before TI's executive leadership, Kilby connected a sliver of semiconductor material to an oscilloscope, pressed a switch, and produced a continuous sine wave. He had invented the monolithic integrated circuit. This proves that every component of an electronic circuit could be fabricated on a single piece of semiconductor material. A feat that earned Kilby the 2000 Nobel Prize in Physics.

> **NARRATOR'S REFLECTION • THE COURAGE TO WALK AWAY FROM GLAMOUR**
>
> *"In technology investing, longevity belongs not to the company that chases the fastest clock speed, but to the company that refuses to play games where the winner must re-bet the entire ranch every twenty-four months."*
>
> Think about how many iconic technology titans of the 1970s, 1980s, and 1990s, from Digital Equipment Corporation and Motorola to Nortel and Nokia, vanished or shrank into irrelevance because they were trapped on a treadmill of short product lifecycles. Texas Instruments survived and compounded for nearly a century because Rich Templeton and Haviv Ilan had the humility to fire their own most glamorous division, mobile phone processors, and bet the company on fifty-cent catalog analog parts that stay in industrial machines for twenty-five years.

The Templeton–Ilan Pivot: Exiting Smartphones to Build the 300mm Analog Fortress

For four decades after Jack Kilby's invention, Texas Instruments competed in almost every corner of electronics: defense radar, minicomputers, scientific calculators, DRAM memory

chips, digital signal processors, and eventually 2G and 3G mobile phone baseband and OMAP application processors. In the early 2000s, if you opened a Nokia or Motorola cellphone, Texas Instruments was the undisputed king inside. Yet by the middle of that decade, Chief Executive Officer Rich Templeton, who led TI as CEO from 2004 to April 2023 and served as Chairman through 2025, spotted a fatal flaw in the economics of digital smartphone chips.

In digital wireless processors, you had to spend billions of dollars every twenty-four months shrinking transistor geometries down the Moore's Law curve at external foundries, only to depend on three or four giant handset customers. Worse still, Templeton saw that handset makers like Apple and Samsung would eventually design their own internal processors or pit suppliers against each other until margins collapsed. Analog and embedded processing chips, by contrast, obeyed completely different economic laws. They ran on mature process nodes between 45 nanometers and 180 nanometers, served more than one hundred thousand industrial and automotive customers, and once designed into an assembly line or car platform, stayed in production for ten to thirty years with virtually zero maintenance R&D.

Working alongside a rising engineering executive named Haviv Ilan. Who had joined Texas Instruments in 1999 when TI acquired wireless startup Butterfly Communications, rose to Chief Operating Officer in 2020, became CEO in April 2023, and assumed the combined role of Chairman and CEO in 2026. Templeton executed one of the boldest capital reallocations in corporate history. Having already sold defense electronics to Raytheon in 1997, exited commodity DRAM, and divested sensors and controls in 2006, TI systematically wound down and exited its multi-billion-dollar wireless baseband and OMAP smartphone processor business between 2008 and 2012.

While Wall Street fretted over the lost handset revenue, Templeton and Ilan deployed that capital counter-cyclically into analog manufacturing during the depth of the 2008–2009 global financial crisis. When German memory chipmaker Qimonda went bankrupt in 2009, Texas Instruments stepped into the liquidation court and bought complete 300-millimeter wafer fabrication toolsets for pennies on the dollar to open RFAB1 in Richardson, Texas. Two years later, in September 2011, TI acquired National Semiconductor for 6.5 billion dollars, adding over 12,000 high-margin power management and precision analog parts.

Rich Templeton spent forty-three years at Texas Instruments instilling the creed that free cash flow per share is the true north of ownership, and when Haviv Ilan became CEO in 2023 and Chairman in 2026 alongside 25-year TI veteran CFO Julie Knecht in August 2026, that cultural continuity remained unbroken.

GLOBAL PARALLEL (EUROPE, USA & ASIA) • THE GREAT DIVERGENCE: NOKIA/INTEL VS. STMICROELECTRONICS, INFINEON & TSMC

When we compare how global semiconductor and electronics leaders navigated the post-2008 era, Texas Instruments' counter-cyclical discipline stands out across three continents:.

- USA. Intel (NASDAQ: INTC) & Motorola: Intel missed mobile, then spent $100B+ trying to catch Taiwan's TSMC (NYSE: TSM) in leading-edge foundry nodes where tools depreciate in 3–4 years. TI exited mobile processors proactively in 2008–2012 and bought 30-year 300mm analog fabs at bankruptcy prices.
- Europe. Infineon (ETR: IFX) & STMicroelectronics (EPA: STM): After Nokia's handset collapse, European peers Infineon and STMicroelectronics also pivoted toward automotive and industrial power. Infineon followed TI's playbook by building 300mm power fabs in Villach (Austria) and Dresden (Germany), though neither matches TI's 80,000-part catalog breadth or 64% Q3 gross margin.
- Asia. TSMC (NYSE: TSM) & India's Tata Electronics: While TSMC dominates sub-5nm digital foundry wafers in Taiwan, Western OEMs realized after 2021 that relying on the Taiwan Strait for $1.00 industrial power chips is an existential supply-chain risk. Fueling both India's new semiconductor assembly initiatives (Tata Electronics, CG Power) and massive customer demand for TI's onshore Texas and Utah fabs.

Anatomy of the 2021–2026 Full Cycle: Boom, Trough, Elliott Activism, and Vindicated Breakout

The five-year arc from 2021 through November 2026 shows how Texas Instruments turns industry downturns into structural market share gains. During the 2021 and 2022 COVID supply crunch, lead times stretched past fifty weeks as OEMs scrambled for 'golden screw' power chips. TI's revenue climbed 26.85 percent in fiscal 2021 to 18,344 million dollars and 9.18 percent in fiscal 2022 to peak at 20,028 million dollars, with gross margin hitting 68.76 percent, operating margin reaching 52.36 percent, and diluted EPS peaking at 9.41 dollars.

Then came the post-pandemic inventory hangover in 2023 and 2024 as customers burned off buffer stockpiles. Revenue fell 12.53 percent in fiscal 2023 to 17,519 million dollars and 10.72 percent in fiscal 2024 to trough at 15,641 million dollars (operating margin 34.21 percent. Diluted EPS 5.20 dollars). Instead of slashing capital spending, Haviv Ilan ramped CapEx to 5,071 million dollars in fiscal 2023, 4,820 million dollars in fiscal 2024, and 4,550 million dollars in fiscal 2025 to complete the long-lead-time 300-millimeter cleanroom shells at RFAB2, LFAB1/LFAB2, and Sherman SM1/SM2.

Because heavy CapEx coincided with trough revenue, standard free cash flow compressed to 1,349 million dollars in 2023 and 1,498 million dollars in 2024. In May 2024, Elliott Investment Management published an open letter disclosing a 2.5 billion dollar stake and urging a dynamic capacity framework to restore free cash flow above 9.00 dollars per share by 2026. At its August 2024 Capital Management update, TI showed how completed cleanroom shells allow management to pace tool hookups flexibly with demand, dropping 2026 CapEx guidance to 2.0–3.0 billion dollars.

Look at the scoreboard on November 6, 2026. As CEO Haviv Ilan emphasized on the October 21, 2026 call, moving dirt to getting wafer output takes four years. Because TI built RFAB2, LFAB1, Sherman SM1, and 4.70 billion dollars of non-perishable inventory during the downturn, TI met the Q3 2026 surge (+28 percent YoY revenue to $6.10B) seamlessly while cutting TTM CapEx to 2,578 million dollars and driving standard free cash flow up 251.8 percent to 7,318 million dollars ($8.40B with CHIPS Act cash).

> **MENTAL MODEL • THE FOUR-YEAR 'DIRT-TO-OUTPUT' CLEANROOM ASYMMETRY**
>
> *"You can wheel a lithography stepper into a finished cleanroom and qualify it in six to nine months. You cannot pour concrete, build ultrapure water plants, and certify a cleanroom shell in less than four years."*
>
> This is the core physical insight that Wall Street missed in 2023–2024 and that Haviv Ilan reiterated on the Q3 2026 earnings call. By separating the multi-year civil construction of cleanroom shells (completed in 2022–2025) from the modular installation of wafer tools (paced dynamically in 2026–2030), Texas Instruments converted fixed construction lag into real-time call-option capacity.

CHAPTER II

Inside the 80,000-Part Catalog — How Analog & Embedded Chips Power AI Racks, Robots & EVs

> *"The excitement I have about industrial is growing very fast. Systems that are being designed by our customers four years later than the COVID peak are much more rich with content. And on the robotic side, per humanoid is very high. Somewhere between $500 to $1,000 of content for TI."*
>
> — HAVIV ILAN, CHAIRMAN & CEO OF TEXAS INSTRUMENTS (Q3 2026 EARNINGS CALL, 21 OCTOBER 2026)

Every digital computation in the modern global economy begins and ends in the physical universe. Think about that for a moment. A thirty-thousand-dollar graphics processing unit inside an artificial intelligence training cluster, a central computer inside an electric vehicle, or a programmable controller on an automated assembly line understands nothing except binary ones and zeros. Yet the physical world in which you and I live does not operate in binary. It consists of continuous analog waves: electrical voltage, amperage, heat, pressure, light, sound, fluid flow, speed, and mechanical torque.

Analog semiconductors are the indispensable translators and power regulators that bridge the physical universe and the digital processor. Before a digital chip can execute a single line of code, analog power management chips must step down high-voltage electricity from the wall or battery, isolate dangerous voltage spikes, and feed a rock-steady sub-one-volt current to the processor core within millivolt tolerances. At the same time, analog signal-chain chips, amplifiers, data converters, sensors, and transceivers, measure real-world temperature, current, and motion, translate those continuous physical signals into digital data streams, and then convert digital commands back into physical motor torque or optical pulses. Paired with Embedded Processing microcontrollers. This act as localized microsecond brains executing real-time control loops, Texas Instruments sells a catalog of more than 80,000 distinct products to over 100,000 customers around the globe.

Let us walk through how Texas Instruments' 20,766 million dollars of trailing twelve-month revenue through September 30, 2026 breaks down across its three reporting segments and

five end markets. Reporting Segment One is the flagship Analog business. This generated 16,815 million dollars over the trailing twelve months, or 80.97 percent of total corporate sales. In the third quarter of 2026, Analog revenue surged 32 percent year-over-year, recovering powerfully from its fiscal 2024 trough of 12,161 million dollars. Analog houses two crown-jewel product families: Power Management. Including DC-to-DC buck and boost converters, low-dropout regulators, gallium nitride power stages, isolated gate drivers, battery management systems, and electronic fuses. And Signal Chain, spanning precision amplifiers, analog-to-digital and digital-to-analog converters, interface transceivers, and reinforced isolation chips.

GLOBAL PARALLEL (USA INDUSTRIAL CATALOGS) • FASTENAL (NASDAQ: FAST) & W.W. GRAINGER (NYSE: GWW) IN SILICON FORM

"Why does an industrial purchasing manager never switch suppliers to save two cents on a fifty-cent fastener or a one-dollar voltage regulator? Because the cost of the part is trivial, but the cost of a halted assembly line is catastrophic."

To understand Texas Instruments' 80,000-part catalog moat, do not compare it to NVIDIA or AMD. Compare it to American industrial distribution titans Fastenal (NASDAQ: FAST) and W.W. Grainger (NYSE: GWW). Fastenal compounds wealth by stocking hundreds of thousands of mission-critical threaded fasteners, bolts, and safety supplies right at the customer's factory floor. In electronics, TI's 80,000 analog and embedded parts are the electrical nuts, bolts, and pressure valves of every circuit board. A single TI current-sense amplifier or buck converter costs $0.50 to $2.00 inside a $300,000 AI server rack or a $100,000 CNC machine. Once an engineer qualifies that TI chip and passes regulatory electromagnetic testing, ripping up the circuit board to save fifteen cents on an unproven rival is economic suicide.

- 80,000+ Active SKUs: An electrical engineer designing a board finds 15 to 25 adjacent power, sensing, and interface parts in one place on TI.com.
- Tiny Fraction of BOM, 100% of Uptime: TI chips represent 0.1% to 1.0% of total machine cost, creating tremendous price inelasticity and 10-to-30-year socket stickiness.
- Zero Obsolescence Inventory: Just as stainless steel bolts do not spoil in a Fastenal bin, a 300mm analog wafer stored in a nitrogen cabinet at RFAB2 stays sellable for a decade.

Reporting Segment Two is Embedded Processing. This generated 3,031 million dollars over the trailing twelve months, or 14.60 percent of corporate sales, growing 21 percent year-over-year in the third quarter of 2026. Embedded Processing builds the C2000 real-time

motor-control microcontrollers, MSPM0 Arm Cortex-M0+ catalog microcontrollers, Sitara industrial application processors, millimeter-wave radar sensors, and wireless connectivity chips. A portfolio soon to be expanded by the Silicon Labs bolt-on acquisition. As CFO Julie Knecht highlighted on the October 21 call, shifting Embedded wafer production from external foundries into TI's 300-millimeter Lehi, Utah fab has expanded Embedded gross margins by roughly 200 basis points per quarter recently and 1,000 basis points year-over-year. Reporting Segment Three is the Other segment at 920 million dollars of trailing revenue, or 4.43 percent of sales. This houses Digital Light Processing micromirror arrays for automotive head-up displays and industrial 3D lithography alongside the iconic TI graphing calculator franchise.

Even more illuminating is the transformation across Texas Instruments' five end markets. Industrial is the number-one vertical at 35 to 40 percent of revenue. Roughly 7.5 to 8.2 billion dollars annualized across ten sub-sectors. And in Q3 2026 it surged more than 40 percent year-over-year and 10 percent quarter-on-quarter to break above its Q3 2022 all-time record. Automotive is the second-largest vertical at 30 to 34 percent of revenue. Roughly 6.5 to 7.0 billion dollars annualized. Growing mid-teens year-over-year and 10 percent sequentially in Q3 as electric and hybrid powertrain systems expanded at twice the overall automotive rate. And Data Center, now TI's third-largest vertical at roughly 13 percent of Q3 revenue, skyrocketed 95 percent year-over-year and 20 percent quarter-on-quarter to reach an annualized run-rate of roughly 2.5 billion dollars. Together, Industrial, Automotive, and Data Center now represent over 80 percent of Texas Instruments' business. Meanwhile, Personal Electronics (10 to 12 percent) and Communications Equipment (4 to 5 percent) are managed selectively for high-return catalog sockets.

Deep Dive 1: Inside a 100-Kilowatt AI Data Center Server Rack (+95% YoY in Q3 2026)

Let us step inside three real-world machines to see how Texas Instruments captures multi-dollar silicon content, starting with an AI data center rack. On the October 21, 2026 call, Morgan Stanley analyst Joe Moore asked CEO Haviv Ilan to unpack why TI's Data Center

revenue surged 95 percent year-over-year and 20 percent quarter-on-quarter to reach roughly 13 percent of company sales (~$2.5B annualized).

Ilan explained the physical reality of high-density AI computing. When a hyperscaler deploys a rack drawing 60 to 120 kilowatts of electrical power, power delivery and thermal telemetry become critical bottlenecks. Electricity enters the rack on a 48-volt DC busbar. Texas Instruments' high-efficiency 48V-to-12V intermediate bus converters and multiphase point-of-load buck stages step voltage down to 0.75V right at the processor socket while delivering hundreds of amps without ripple.

Across the blade, TI high-current electronic fuses (eFuses) and hot-swap controllers prevent electrical arcing. Meanwhile, INA-series current-sense amplifiers and TMP-series temperature sensors monitor every PCIe lane, optical module, and liquid-cooling manifold, and isolated gate drivers spin cooling pumps and fans. As Haviv Ilan told Joe Moore verbatim: 'We reviewed it earlier this week. Also, we have about 10,000 different GPNs or generic part numbers that we are selling into this market. Also, it is growing every quarter. We also pay attention to the more application-specific sockets. We have developed the technology for that. Also, it is now early ramping in Sherman'.

> **NARRATOR'S REFLECTION • 10,000 PART NUMBERS IN THE AI DATA CENTER**
>
> *"You can swap out which GPU or custom ASIC sits in the center of the server blade, but every single rack still needs 48-volt power conversion, hot-swap protection, liquid-cooling pump drives, and current telemetry."*
>
> Notice the beauty of selling 10,000 catalog and application-specific analog part numbers into the data center: Texas Instruments is completely agnostic to whether NVIDIA, AMD, Broadcom, or an internal cloud ASIC wins the digital compute socket. Every kilowatt of added rack power density automatically multiplies the number of TI power stages, eFuses, and thermal sensors on the board.

Deep Dive 2 & 3: Six-Axis Factory Robots, $1,000 Humanoids, and 800-Volt Electric Vehicles

Now walk with me from the hyperscale data center onto a modern automated factory floor, where Texas Instruments' Industrial end market surged more than 40 percent year-over-year in the third quarter of 2026 to break above its Q3 2022 peak for the first time in four years. Why did Industrial revenue reach a new all-time high even while several industrial sub-sectors are still early in their cyclical recovery? Because every four-year design generation of factory machinery packs 20 to 40 percent more analog and embedded silicon inside.

Picture a six-axis articulated robotic arm welding car bodies or handling semiconductor wafers with sub-millimeter precision. Inside every single joint of that robot sits a closed-loop motor drive subsystem anchored by a Texas Instruments C2000 real-time microcontroller. That C2000 chip samples the rotor angle, computes complex field-oriented torque math, and updates pulse-width modulation switching signals in less than one microsecond. Ten times faster than a standard processor. Surrounding that C2000 controller on the same circuit board are dozens of TI analog chips: reinforced capacitive digital isolators and isolated delta-sigma modulators protecting low-voltage logic from 400-to-800-volt motor currents, gallium nitride power stages that eliminate bulky heatsinks, industrial Ethernet transceivers, and 60-gigahertz millimeter-wave radar sensors on the robot's exterior shell that create an invisible 3D safety curtain whenever a human worker approaches.

Even more exciting is what happens when factory automation evolves into autonomous humanoid robots. On the October 21, 2026 call, Haviv Ilan revealed that while traditional robotics is already growing, 2027 will mark a wave of commercial production ramps in humanoid robots, and because a humanoid packs thirty to fifty individual motor actuators, tactile sensors, radar eyes, and battery management systems into a single frame, Texas Instruments captures between 500 dollars and 1,000 dollars of semiconductor content per humanoid robot.

Finally, look under the chassis of a modern battery electric or plug-in hybrid vehicle. A sub-sector that Head of Investor Relations Mike Beckman confirmed grew roughly twice as fast as TI's overall automotive business in Q3 2026, pulling total Automotive revenue up mid-

teens year-over-year and 10 percent sequentially. Inside an 800-volt lithium-ion battery pack sit 96 to 192 cell groups that must be monitored within 1.5 millivolts from minus 40 to plus 125 degrees Celsius. Texas Instruments supplies the BQ796xx family of daisy-chained and wireless battery management monitors, capacitive isolation barriers, pyro-fuse battery disconnect drivers, and smart high-side switches across modern zonal body architectures in both Detroit and Shenzhen, locking in recurring sockets for the entire seven-to-ten-year production life of the vehicle.

GLOBAL PARALLEL (EUROPE & ASIA) • STMICROELECTRONICS, INFINEON & CHINA EV ARCHITECTURES

In global automotive and industrial robotics, European peers Infineon (ETR: IFX) and STMicroelectronics (EPA: STM) and Japan's Renesas compete fiercely in high-voltage silicon-carbide traction inverters and microcontrollers. Yet when Chinese and Western EV makers re-architected their vehicles around 48V/800V zonal architectures and advanced ADAS in 2025–2026, TI's unmatched catalog breadth allowed engineers to source battery monitors, isolated gate drivers, FPD-Link camera serializers, and CAN/LIN transceivers from a single vendor. Explaining why TI's China EV revenue grew faster than the company average in Q3 2026.

CHAPTER III

The 300mm Wafer Physics, the TI.com Direct Revolution & the $100 Revenue Waterfall

> *"A 300-millimeter wafer has 2.25 times the surface area of a 200-millimeter wafer and yields roughly 2.3 times as many dies. Yet costs only about 1.35 times as much to process. Delivering a structural 40 percent cost advantage on every unpackaged chip."*
>
> — TEXAS INSTRUMENTS CAPITAL MANAGEMENT & 300MM MANUFACTURING PHYSICS

If you ask a generalist investor why Texas Instruments earns a 64 percent gross margin and a 47 percent operating margin while selling chips that often cost less than a cup of coffee, they will usually guess that TI holds secret patents that nobody else can copy. Patents and circuit design talent certainly matter, but two deeper operational revolutions executed over the past fifteen years truly separate Texas Instruments' unit economics from every other analog chipmaker on the planet. The first is the dismantling of the traditional third-party electronics distribution middleman in favor of direct customer relationships and TI.com. The second is the multi-decade migration of analog and embedded wafer manufacturing from legacy 200-millimeter wafers to captive 300-millimeter cleanrooms paired with internal assembly and test mega-sites.

Let us examine the distribution revolution first. That is because it is one of the most underrated strategic masterstrokes in modern corporate history. For half a century, the global analog semiconductor industry relied on giant third-party electronics distributors, companies like Avnet, Arrow Electronics, and Taiwan's WPG Holdings, to reach the long tail of one hundred thousand industrial and commercial customers. Under that old model, roughly two-thirds of an analog chipmaker's revenue flowed through distributors. Those middlemen demanded a 10 to 15 percent gross margin cut through complex 'ship-and-debit' price rebates, hid the identity and real-time demand of end customers behind a wall of channel fog, and violently amplified semiconductor boom-and-bust cycles by hoarding inventory during shortages and canceling orders en masse during downturns.

Between late 2019 and the end of 2020, Texas Instruments did what no other major analog supplier dared to do: it terminated its franchise distribution agreements with Avnet, WPG,

and WT Microelectronics, retaining only Arrow and catalog e-tailers DigiKey and Mouser while migrating more than 70 to 75 percent of its total revenue directly onto TI.com and direct corporate accounts. Today, when an electrical engineer in Munich, Detroit, Bengaluru, or Austin designs a circuit board, they open TI's online WEBENCH simulation tool, verify thermal and loop stability in minutes, check live factory reel inventory down to the exact unit on TI.com, pay in over thirty local currencies, and receive cut-tape or full-reel shipments within 48 hours straight from TI's automated distribution hubs. TI recaptured the distributor margin spread directly into its own income statement, gained an unclouded real-time radar into end-customer demand, and shifted inventory management onto its own balance sheet, where 4.70 billion dollars of non-perishable die bank and finished reels sit ready for immediate shipment.

> **GLOBAL PARALLEL (USA DIRECT CHANNEL) • APPLE STORES, TESLA DIRECT & TI.COM — OWNING THE CUSTOMER RELATIONSHIP**
>
> *"Whenever a manufacturer eliminates a 12% middleman markup while simultaneously improving delivery speed and capturing real-time customer telemetry, the moat widens on both sides of the ledger."*
>
> When Texas Instruments cut off Avnet and WPG in 2019–2020, Wall Street analysts warned that TI would lose tens of thousands of small industrial customers who relied on distributor field engineers. The exact opposite happened. Mirroring Apple's (NASDAQ: AAPL) launch of direct retail and online stores or Fastenal's automated onsite vending machines. By investing in WEBENCH automated design software, application notes, and 48-hour direct TI.com fulfillment, TI turned its website into the automatic starting point for global electrical engineers while keeping the distributor's gross margin cut for TXN shareholders.

Now combine that direct-to-customer engine with the inexorable geometry of a 300-millimeter silicon wafer. While digital processor designers like Apple and NVIDIA moved to 300-millimeter (12-inch) wafers at TSMC twenty years ago, almost the entire global analog industry stayed on older 150-millimeter (6-inch) and 200-millimeter (8-inch) wafers. Why? Because an analog chip is physically tiny and sells for fifty cents to two dollars. Filling a single 300-millimeter fab requires shipping billions of units across tens of thousands of part numbers. Because the area of a circle equals pi times the radius squared, increasing wafer diameter by 1.5 times from 200 millimeters to 300 millimeters expands total silicon surface

area by 2.25 times, from 31,416 square millimeters to 70,686 square millimeters, and because less perimeter percentage is lost to curved edge waste, a 300-millimeter wafer yields roughly 2.3 to 2.4 times as many usable chips. Yet running that 300-millimeter wafer through photolithography, ion implantation, and etch costs only about 1.35 to 1.40 times as much as a 200-millimeter wafer, cutting unpackaged die cost by roughly 40 percent and lifting packaged chip gross margin by roughly 800 basis points.

A Physical Tour of TI's 300mm Wafer Fabs and Internal Assembly Mega-Sites

Let us tour Texas Instruments' physical manufacturing footprint as it operates in November 2026. In Dallas, DMOS6 operates as a mature, heavily depreciated 300-millimeter analog and power fab generating massive cash flow with minimal ongoing CapEx. Just north in Richardson, Texas sits RFAB1, opened in 2009 as the world's first 300-millimeter analog fab, connected by an enclosed cleanroom tunnel directly to RFAB2. This opened in September 2022 and is scaling output across thousands of catalog parts. Overhead automated carriers shuttle 300-millimeter wafer pods seamlessly between RFAB1 and RFAB2, sharing engineers, chemicals, and ultrapure water plants.

Thirty miles north up U.S. Highway 75 in Sherman, Texas sits the centerpiece of TI's multi-decade vision: the four-fab Sherman mega-campus (SM1, SM2, SM3, and SM4). As Factory Manager Mike Haggerty and CEO Haviv Ilan showcased when SM1 began initial production in December 2025, SM1 is already fabricating 300-millimeter analog and data center power wafers in 2026 while the exterior shell of SM2 stands completed beside it. Meanwhile, in Lehi, Utah sits LFAB1, acquired from Micron for 900 million dollars in 2021. That is internalizing Embedded Processing wafers and lifting Embedded gross margins by 1,000 basis points year-over-year, right next to the LFAB2 cleanroom shell reaching completion at year-end 2026.

On the back end, when analysts on the October 21, 2026 call noted that third-party OSAT packaging providers in Asia are suffering from capacity bottlenecks and extending lead times, Haviv Ilan explained why TI was unfazed: TI had just ramped its new automated assembly and test mega-site in Melaka and Kuala Lumpur, Malaysia, now TI's largest back-

end facility, working alongside its captive plants in Clark, Philippines and Mexico to push internal assembly and test share toward 85 to 90 percent.

> **GLOBAL PARALLEL (EUROPE & INDIA/ASIA) • ASML DUV VS. EUV & INDIA'S OSAT EXPANSION (TATA, CG POWER, KAYNES)**
>
> Two international comparisons highlight the brilliance of TI's front-end and back-end manufacturing architecture:.
>
> - Netherlands. ASML (NASDAQ: ASML) Tool Economics: Leading-edge digital fabs must buy ASML's $380M High-NA EUV lithography systems. TI's 300mm fabs in Sherman and Lehi run 45nm to 130nm BCD and precision analog processes using mature, battle-tested DUV immersion and dry steppers that cost a fraction of EUV and run for three decades.
> - India & Southeast Asia. OSAT Bottlenecks (Tata Electronics, CG Power, Kaynes): While Western fabless chipmakers are scrambling in 2026 to secure packaging slots across Taiwan and newly emerging Indian OSAT plants (such as Tata Electronics, CG Power's Sanand plant, and Kaynes Semicon), TI's captive automated assembly mega-site in Malaysia already insulates its 100,000 customers from back-end shortages.

Walking $100 of Revenue Down to $35.24 of TTM Free Cash Flow (and $48.15 in Q3 2026)

To see how 300-millimeter wafer physics and direct TI.com distribution turn into spendable owner cash, let us walk 100 dollars of revenue down Texas Instruments' financial statements across three snapshots: the fiscal 2024 trough ($15,641M sales), the trailing twelve months ended September 30, 2026 ($20,766M sales), and the third-quarter 2026 run-rate ($6,100M quarter, or $24.40B annualized).

Start at the top line with 100 dollars of revenue. At the fiscal 2024 trough, cost of goods sold consumed 41 dollars and 86 cents. That leaves 58 dollars and 14 cents of gross profit. Over the trailing twelve months ended September 2026, cost of goods sold fell to 39 dollars and 76 cents (including $2,180M of factory depreciation). That lifted gross profit to 60 dollars and 24 cents. In the third quarter of 2026 alone, as higher wafer starts flowed through RFAB2, LFAB1, and Sherman SM1 at nearly 90 percent incremental margin, cost of goods sold dropped to 36 dollars. That leaves 64 dollars of gross profit per 100 dollars of sales.

Next, subtract operating expenses (R&D and SG&A). Because a catalog chip designed ten years ago requires zero ongoing redesign, operating expenses per 100 dollars of sales fell from 23 dollars and 93 cents in fiscal 2024 to 19 dollars and 15 cents TTM, and down to 17 dollars in Q3 2026. Operating Income (EBIT) per 100 dollars of sales surged from 34 dollars and 21 cents in fiscal 2024 to 41 dollars and 9 cents TTM, and to 47 dollars in Q3 2026.

After subtracting net interest and taxes (~13 percent effective rate), GAAP net income equals 34 dollars and 6 cents TTM and 39 dollars and 34 cents in Q3. Adding back non-cash depreciation ($2,263M TTM), stock-based compensation ($412M TTM), and working capital items lifts Operating Cash Flow to 47 dollars and 66 cents per 100 dollars of TTM revenue ($9,896M) and 55 dollars and 74 cents in Q3 ($3,400M). Subtracting Gross Capital Expenditures. Which dropped from 30 dollars and 82 cents per 100 dollars of sales in FY24 to 12 dollars and 41 cents TTM and just 7 dollars and 59 cents in Q3. Leaves 35 dollars and 24 cents of pure standard free cash flow per 100 dollars of TTM sales, and 48 dollars and 15 cents in Q3 2026.

> **MENTAL MODEL • THE DOUBLE-LEVERAGE SCISSORS OF AN IDM HARVEST**
>
> Look at the two blades of the scissors between FY2024 and Q3 2026: Operating Cash Flow per $100 of sales rose from $40.39 to $55.74 as 300mm fabs filled up. Meanwhile, CapEx per $100 of sales fell from $30.82 to $7.59 as cleanroom construction finished. When rising operating cash flow meets falling capital intensity, free cash flow margin does not merely grow. It multiplies fivefold from 9.58% to 48.15%.

CHAPTER IV

The Four Pillars of the Analog Moat, Geopolitical Pricing Power & Competitor Breakdown

> *"In the area of foundational chips, specifically analog, there has not been a lot of addition into capacity in the last four years. So especially if you do care about geopolitically dependable capacity, meaning factories outside of China and Taiwan, we are looking at the customer's demand and what type of supply they want to guarantee from TI. Also, that has a pricing element to it."*
>
> — HAVIV ILAN, CHAIRMAN & CEO OF TEXAS INSTRUMENTS (Q3 2026 EARNINGS CALL, 21 OCTOBER 2026)

Whenever you open a Texas Instruments capital management presentation or listen to Haviv Ilan and Julie Knecht address shareholders, they judge every decision against four interlocking pillars of competitive advantage. Pillar One is Manufacturing and Technology. That means owning internal 300-millimeter wafer fabrication, internal back-end assembly and test, and proprietary process technologies. Pillar Two is a Broad Product Portfolio of more than 80,000 catalog analog and embedded parts. Pillar Three is the Reach of Market Channels, serving over 100,000 customers directly through TI.com and internal field engineers. Pillar Four is Diverse and Long-Lived Positions across industrial, automotive, and data center systems lasting ten to thirty years.

What makes Texas Instruments unique in global semiconductors is the multiplicative force of all four pillars working together at 300-millimeter scale. A fabless rival can design a good voltage regulator. However, it must buy expensive external foundry wafers and pay a distributor 15 percent to reach customers. Even large Western competitors only match TI on one or two pillars. Analog Devices boasts strong precision signal-chain engineering from its Linear Technology and Maxim deals. NXP Semiconductors and Germany's Infineon hold deep automotive relationships. Yet none of them combines an 80,000-part catalog with a 70-percent-plus direct e-commerce channel and six 300-millimeter wafer fabs in the United States.

NARRATOR'S REFLECTION • THE FIVE-YEAR GAAP DEPRECIATION VS. THIRTY-YEAR PHYSICAL LIFE SECRET

"An accountant depreciates a 300mm analog lithography stepper to zero in sixty months. An electrical engineer runs that exact same machine in the cleanroom for thirty years."

Here is the single most important accounting insight you and I must hold onto when analyzing Texas Instruments. Under US GAAP rules, TI depreciates wafer fabrication tools over just five years. That is why trailing twelve-month depreciation and amortization hit $2,263 million in Q3 2026. But physically, a 300mm stepper or ion implanter running 45nm to 130nm analog chips does not wear out in five years. It runs for 25 to 30 years (just as DMOS6 and RFAB1 have proven). As the 2022–2025 tool cohorts roll off their five-year GAAP depreciation schedules between 2027 and 2030, billions of dollars in non-cash depreciation expense will vanish from COGS while the machines keep printing 68%+ gross margin silicon.

On the October 21, 2026 third-quarter earnings call, a powerful new weapon born from those four pillars took center stage: geopolitically dependable capacity and 2027 annual contract pricing power. During the Q&A session, JPMorgan analyst Harlan Sur asked CEO Haviv Ilan whether tightening supply was enabling Texas Instruments to negotiate positive price increases across Analog and Embedded Processing for 2027. For decades prior to 2020, analog chip prices gently declined by one to two percent per year as volume scaled.

Ilan's response to Harlan Sur explained why the post-2026 market is structurally different. Since the COVID peak four years ago in 2022, virtually zero new foundational analog wafer capacity has been built outside of mainland China and Taiwan. Meanwhile, external foundries like TSMC devoted almost all new capital and cleanrooms to digital AI logic and CoWoS packaging rather than mature analog nodes. As Industrial (+40 percent year-over-year), Data Center (+95 percent year-over-year), and Automotive (+mid-teens year-over-year) orders accelerated in the second half of 2026, supply tightened and lead times ticked up.

For Western and global industrial, automotive, and cloud OEMs that require supply-chain resilience outside the Taiwan Strait and mainland China, Texas Instruments' 300-millimeter fabs in Richardson, Sherman, and Lehi, paired with internal assembly plants in Malaysia, the Philippines, and Mexico, are the only large-scale sanctuary on earth. Ilan confirmed that after flat pricing in the first half of 2026, TI captured positive price increases in Analog

during the second half of 2026 on urgent volume, and is now embedding positive pricing into annual 2027 supply-guarantee agreements.

> **GLOBAL PARALLEL (CHINA, TAIWAN & WESTERN PEERS) • CHINA'S SILERGY & SG MICRO VS. TI'S HIGH-SPEC EV & INDUSTRIAL FORTRESS**
>
> One of the most frequent bear questions is whether Beijing's push for indigenous semiconductors. It was led by Chinese analog houses like Silergy, SG Micro, and 3PEAK, will commoditize Texas Instruments' catalog. The Q3 2026 evidence tells a remarkably clear story:.
>
> - Where China Competes: Domestic Chinese analog firms compete aggressively on price in low-voltage consumer electronics, small home appliances, and basic 200mm catalog regulators.
> - Why TI's China Revenue Outpaced the Company Average in Q3 2026: As Haviv Ilan and Mike Beckman disclosed on the October 21 call, TI's China business grew faster than overall corporate revenue (+28% YoY) in Q3 2026. It was led by Chinese electric vehicle and hybrid powertrains. Because Chinese EV leaders (such as BYD, Xiaomi, and NIO) export globally to Europe and Latin America and pack 800V battery management and ADAS into their cars, they require TI's globally certified AEC-Q100 and ISO 26262 chips.
> - The 40% 300mm Cost Shield: Even when TI chooses to defend a standard industrial socket in Asia, its 40% unpackaged die-cost advantage on 300mm wafers allows TI to match local 200mm foundry-based competitors on price while still earning a 60%+ gross margin.

Head-to-Head Competitor Breakdown: Why No Peer Can Replicate TI's Four Pillars Before 2032

Suppose you were appointed chief executive of a rival analog chipmaker today and ordered to replicate Texas Instruments' 300-millimeter cost structure and direct channel moat. How long would it take? First, permitting, pouring concrete, erecting the cleanroom shell, and certifying ultrapure water and gas plants for a greenfield 300-millimeter fab requires four full years (taking you to late 2030) and five to eight billion dollars of capital before a single wafer exits the line. Second, unlike a digital chipmaker that qualifies three processor designs to fill a fab, an analog rival must port and obtain customer qualification for tens of thousands of catalog part numbers across automotive and industrial safety standards. That takes another three to five years. Because TI began this journey in 2009 with RFAB1 and completed its mega-shells in 2022–2025, TI holds a seven-to-ten-year lead.

We see the proof of that lead by comparing Texas Instruments head-to-head against its four primary U.S.-listed peers on verified November 6, 2026 numbers. Start with Analog Devices (NASDAQ: ADI) at 365 dollars and 71 cents (177.21 billion dollar market cap). Through its acquisitions of Linear Technology and Maxim Integrated, ADI is strong in precision signal chain. That generated 13.88 billion dollars of trailing sales with a 60.39 percent gross margin and a 35.89 percent operating margin. Yet because ADI outsources wafers to TSMC and carries large acquisition goodwill, ADI's return on invested capital is just 10.25 percent and its ROE is 12.23 percent. That is less than half of Texas Instruments' 22.04 percent ROIC and less than a third of TI's 39.50 percent ROE.

Next, look at NXP Semiconductors (NASDAQ: NXPI) at 247 dollars and 75 cents (62.45 billion dollar market cap) and Microchip Technology (NASDAQ: MCHP) at 58 dollars and 78 cents (31.92 billion dollar market cap). NXP focuses on automotive processors and radar across 13.80 billion dollars of sales. It posted a 55.68 percent gross margin, a 29.92 percent operating margin, and a 14.63 percent ROIC. Microchip competes against TI in microcontrollers and analog peripherals across 5.60 billion dollars of sales, but suffered a severe drawdown after over-leveraging its balance sheet. Microchip's operating margin fell to 16.23–21.57 percent, its ROIC dropped to 6.89 percent, and its stock sits 44.50 percent below its 52-week high.

Finally, examine ON Semiconductor (NASDAQ: ON) at 63 dollars and 56 cents (24.74 billion dollar market cap). When EV silicon-carbide pricing came under pressure, ON's gross margin fell to 38.13 percent, its operating margin dropped to 19.23 percent, its ROIC sank to 6.02 percent, and its shares fell 52.89 percent from their 52-week high. Meanwhile, in Embedded Processing, Texas Instruments is integrating its bolt-on Silicon Labs assets ($34 million of H1 2026 integration charges) and porting those foundry wafers into its Lehi LFAB1 and LFAB2 300-millimeter cleanrooms for an 800-to-1,000-basis-point gross margin uplift.

MENTAL MODEL • THE EXTERNAL LEARNING HUB FOR SELF-DIRECTED OWNERS

In Chapter 4 of the companion Web Research Dossier, we compiled ten primary-source links so you can verify every claim firsthand: the full October 21, 2026 Q3 earnings call transcript, the EE Times and YouTube cleanroom tours of the Sherman SM1 300mm fab with CEO Haviv Ilan and Factory Manager Mike Haggerty, the audited 2025 Form 10-K and Q2 2026 Form 10-Q filings, Drillr's 300mm cycle monograph, and ThesisCheck's forensic filing teardown.

CHAPTER V

The Six Leading KPIs, Verbatim Q3 2026 Management Commentary & Walk-the-Talk Audit

> *"Gross profit in the quarter was $3.9 billion or 64% of revenue. Sequentially, gross profit margin increased 280 basis points. Operating profit was $2.9 billion in the quarter, or 47% of revenue, and was up 73% from the year-ago quarter. Free cash flow on a trailing 12-month basis was $8.4 billion, up from $2.4 billion in the third quarter of 2025."*
>
> — JULIE C. KNECHT, CHIEF FINANCIAL OFFICER OF TEXAS INSTRUMENTS (Q3 2026 EARNINGS CALL, 21 OCTOBER 2026)

If you try to invest in an integrated semiconductor manufacturer like Texas Instruments by waiting for backward-looking twelve-month GAAP earnings per share to look pretty on a Bloomberg or Finviz screen, you will arrive late at every single cyclical turning point. Because Texas Instruments owns and operates its own 300-millimeter front-end wafer fabs and back-end assembly plants, its true economic trajectory is governed by six leading operational indicators that turn months before headline GAAP ratios catch up.

What are those six leading indicators that you and I must track every quarter? First is sequential revenue momentum and fourth-quarter guidance relative to historical seasonal norms. Second is gross margin and incremental sequential revenue fall-through excluding depreciation. Third is balance-sheet inventory days and die-bank readiness. Fourth is capital expenditure intensity relative to completed cleanroom shells. Fifth is United States CHIPS and Science Act tax credit and grant cash receipts. And sixth is free cash flow per share and dividend coverage. In the third quarter of 2026, all six leading indicators flipped decisively from headwind to tailwind simultaneously.

Look first at sequential revenue momentum and fourth-quarter seasonality. Not only did third-quarter 2026 revenue jump 28 percent year-over-year and 11 percent quarter-on-quarter to reach 6.10 billion dollars, but management guided fourth-quarter 2026 revenue to a range of 5.80 billion to 6.30 billion dollars, with a midpoint of 6.05 billion dollars and diluted EPS of 2.45 to 2.79 dollars. During the Q&A session on October 21, Bernstein analyst Stacy Rasgon pointed out the obvious historical anomaly: in normal years, Texas

Instruments' fourth calendar quarter drops five to eight percent sequentially as holiday consumer builds wrap up and industrial plants slow for year-end holidays. Guiding fourth-quarter revenue essentially flat at the midpoint, and up three percent sequentially at the top end, represents TI's strongest winter seasonal outlook since the 2021 pandemic boom, powered by secular Data Center expansion (+95 percent year-over-year) and Industrial breaking above its 2022 record (+40 percent year-over-year).

Now look at KPI Number Two: gross margin and incremental fall-through. The October 21, 2026 call marked one of the debut quarterly briefings for Chief Financial Officer Julie Knecht, a twenty-five-year Texas Instruments finance veteran who succeeded retiring CFO Rafael Lizardi on August 1, 2026. A transition that TI's June 2, 2026 Form 8-K confirmed was a planned retirement unrelated to financial results or internal controls. Knecht walked listeners through the exact margin math: third-quarter gross profit hit 3.90 billion dollars, or 64.0 percent of revenue, up 280 basis points sequentially. On 600 million dollars of sequential revenue growth, gross profit rose by more than 530 million dollars. An incremental gross margin fall-through approaching 90 percent. While Knecht advised analysts to continue modeling 75 to 85 percent long-term revenue fall-through excluding depreciation, she confirmed that higher factory loadings at RFAB2, LFAB1, and Sherman SM1, richer Industrial mix, and second-half analog pricing gains all contributed to the surge.

> **NARRATOR'S REFLECTION • WHY 194 DAYS OF ANALOG INVENTORY IS A WEAPON, NOT A LIABILITY**
>
> *"In smartphones or fashion retail, six months of inventory is rot waiting for a write-off. In catalog analog silicon stored in nitrogen cabinets, 194 days of inventory is the ultimate market-share bazooka."*
>
> Generalist analysts often panic when they see $4.70 billion of inventory (194 days, down 2 days QoQ) on Texas Instruments' balance sheet. Listen to what actually happened in Q3 2026 when Wells Fargo's Joseph Quatrochi and Truist's William Stein asked about customer inventory behavior: automakers and industrial OEMs had cut their own component shelves too lean and suddenly needed immediate 'golden screw' shipments to build EVs and factory robots. While fab-lite peers quoted extending lead times, TI shipped immediately from its 194-day die bank and finished-goods buffer, taking permanent socket share.

Equally impressive was the operational turn in Embedded Processing. When Wolfe Research analyst Chris Caso asked about the margin trajectory of the Lehi, Utah 300-millimeter fabs, Haviv Ilan and Julie Knecht revealed that migrating Embedded Processing wafers out of external foundries and into Lehi LFAB1 has expanded Embedded gross margins by roughly 200 basis points per quarter over recent quarters, or approximately 1,000 basis points, ten full percentage points, year-over-year. With the adjacent LFAB2 cleanroom shell reaching completion at the end of 2026, TI is positioned to repeat that exact internal wafer margin uplift across the incoming Silicon Labs portfolio.

CapEx Modulation, the $1.60B CHIPS Act Windfall, and the CEO Walk-the-Talk Scorecard

Let us examine KPIs Four, Five, and Six: Capital Expenditure Intensity, CHIPS Act Cash Receipts, and Free Cash Flow per Share. In the third quarter of 2026, Texas Instruments spent just 463 million dollars on capital expenditures. Only 7.6 percent of quarterly revenue. Bringing trailing twelve-month CapEx down to 2,578 million dollars, or 12.4 percent of trailing sales. Compare that to fiscal 2024, when CapEx consumed 4,820 million dollars, or 30.8 percent of sales. On the October 21 call, Haviv Ilan confirmed that full-year 2026 CapEx remains on track for the guided 2.0 billion to 3.0 billion dollar range, leaning toward the upper half as back-end assembly tools in Malaysia and front-end wafer tools in Sherman and Lehi are delivered to support customer demand.

Now pair that CapEx step-down with KPI Number Five: CHIPS Act cash inflows. As Julie Knecht highlighted verbatim on the call, Texas Instruments received 1.60 billion dollars of cash from United States CHIPS and Science Act incentives over the trailing twelve months ended September 30, 2026. Combining the 25 percent advanced manufacturing investment tax credit with direct grant disbursements. Think about that arithmetic: against 2,578 million dollars of gross capital expenditures over the past year, the U.S. Treasury reimbursed 1,600 million dollars in cash. That leaves TI's net cash capital outlay at less than one billion dollars for an entire year of 300-millimeter expansion. That is why CHIPS-inclusive trailing twelve-month free cash flow soared 3.5-fold from 2.40 billion dollars in Q3 2025 to 8.40

billion dollars in Q3 2026 (with standard free cash flow excluding direct grants up 251.8 percent to 7,318 million dollars, or 7.99 dollars per share).

How does management's multi-year execution grade out when you and I audit their 2022–2025 promises against delivered reality in late 2026? Across four major commitments, our forensic Walk-the-Talk scorecard awards management an institutional grade of A-minus. First, on the counter-cyclical 300-millimeter cleanroom buildout: management promised to invest roughly 5 billion dollars a year through the 2023–2025 downturn so cleanrooms would be ready for the next upcycle, and today RFAB2, LFAB1, Sherman SM1, and the Malaysia assembly mega-site are live and powering +28 percent year-over-year revenue growth. Second, on the post-Elliott dynamic capacity framework announced in August 2024: management promised to step 2026 CapEx down to 2.0–3.0 billion dollars and restore free cash flow toward 8 to 9 dollars per share, and TTM CapEx sits at 2.58 billion dollars while FCF per share has reached 7.99 dollars standard and roughly 9.15 dollars with CHIPS cash.

Third, on Embedded Processing margin recovery: management promised that Lehi LFAB1 internalization would close the margin gap with Analog, and Embedded gross margins have expanded 1,000 basis points year-over-year. Fourth, on shareholder cash returns: TI raised its dividend by 7 percent in September 2026 to 1.52 dollars per quarter (6.08 dollars annualized) for its 23rd consecutive annual increase, returning 5.80 billion dollars TTM (5.17 billion in dividends and 636 million in buybacks). The only minor blemish. This keeps the grade at A-minus rather than A-plus, is that throttled buybacks during the peak CapEx years allowed employee stock option exercises (812 million dollars TTM) to nudge diluted shares up by roughly 0.9 percent from 912 million to between 914 million and 920 million shares, a trend we expect CFO Julie Knecht to reverse as excess cash builds in 2027.

GLOBAL PARALLEL (USA & EUROPE CHIPS ACT EXECUTION) • WHY TI COLLECTED $1.6B IN CHIPS CASH WHILE INTEL & OTHERS STALLED

The US CHIPS and Science Act and the European Chips Act offered tens of billions in subsidies. Yet payouts are tied to actual construction and production milestones. While Intel (NASDAQ: INTC) delayed fab projects in Ohio and Europe and TSMC (NYSE: TSM) faced initial skilled-labor delays in Arizona, Texas Instruments executed like clockwork. Completing shells in Richardson, Lehi, and Sherman on schedule and collecting $1.60 billion in verified CHIPS Act cash over the trailing twelve months ended Q3 2026.

CHAPTER VI

Five-Year & TTM Financial Forensics — Income Statement, Cash Conversion & 10-Q/10-K Balance Sheet Teardown

> *"Accounting earnings tell you what the tax authorities and depreciation schedules allow you to report. Cash flow from operations minus maintenance capital tells you what the business actually earned."*
>
> — INSTITUTIONAL FORENSIC ACCOUNTING MAXIM

Now it is time for you and me to roll up our sleeves, open Texas Instruments' audited Form 10-K filings, second-quarter Form 10-Q footnotes, and third-quarter 2026 financial statements, and conduct a line-by-line forensic inspection across the entire 2021–2026 cycle. Because listeners enjoying the audiobook edition do not have visual spreadsheets in front of them, I am going to narrate the exact trajectory of revenue, margins, cash conversion, and balance-sheet leverage so you can hear the rhythm of the business.

Begin with the top line across our six audited windows: fiscal 2021, fiscal 2022, fiscal 2023, fiscal 2024, fiscal 2025, and the trailing twelve months ended September 30, 2026 (plus the third-quarter 2026 annualized run-rate). In fiscal 2021, total revenue surged 26.85 percent to 18,344 million dollars. It was led by Analog at 14,050 million dollars and Embedded Processing at 3,049 million dollars. In fiscal 2022, revenue rose another 9.18 percent to reach the pandemic peak of 20,028 million dollars, with Analog at 15,359 million dollars and Embedded at 3,261 million dollars. Then came the two-year post-pandemic destocking contraction: revenue fell 12.53 percent in fiscal 2023 to 17,519 million dollars (Analog 13,040 million. Embedded 3,368 million) and dropped 10.72 percent in fiscal 2024 to trough at 15,641 million dollars (Analog 12,161 million. Embedded 2,533 million).

In fiscal 2025, the cyclical turn began. That lifted total revenue 13.05 percent to 17,682 million dollars (Analog 14,006 million. Embedded 2,697 million). And over the trailing twelve months ended September 30, 2026, total revenue jumped 20.27 percent to establish a new all-time record of 20,766 million dollars. With Analog hitting a record 16,815 million dollars, Embedded rebounding to 3,031 million dollars, and Other contributing 920 million dollars. Even more telling, third-quarter 2026 revenue of 6,100 million dollars (+28 percent

year-over-year and +11 percent sequentially) puts Texas Instruments on a 24,400 million dollar annualized revenue run-rate entering the final quarter of the year.

Now listen to how Texas Instruments' profitability held up through that violent cyclical swing. At the 2021 and 2022 peak, gross margin stood at 67.47 percent and 68.76 percent. Meanwhile, operating margin hit 49.18 percent and 52.36 percent. That generated 10,487 million dollars of peak operating income and 9.41 dollars of diluted EPS in 2022. When revenue fell by 4.39 billion dollars into the 2024 trough and newly opened 300-millimeter fabs added more than a billion dollars of fixed annual depreciation, did Texas Instruments fall into losses like commodity chipmakers of the past? Not even close. Gross margin never dipped below 58.14 percent in 2024 or 57.02 percent in 2025, and operating margin bottomed at an enviable 34.21 percent in 2024 (5,350 million dollars of EBIT and 5.20 dollars of diluted EPS). And the moment revenue broke out in the third quarter of 2026, gross margin snapped right back to 64.00 percent (60.24 percent TTM) and operating margin leaped to 47.00 percent (2,900 million dollars of quarterly EBIT, or 11,600 million dollars annualized). That drove Q3 diluted EPS to 2.62 dollars (10.48 dollars annualized).

> **MENTAL MODEL • A 34.2% TROUGH OPERATING MARGIN IS THE ULTIMATE MOAT PROOF**
>
> *"Show me a company's operating margin at the bottom of the worst inventory correction in fifteen years, and I will show you the true width of its moat."*
>
> Most industrial and technology companies dream of earning a 34% operating margin at the peak of an economic boom. Texas Instruments earned a 34.21% operating margin at the absolute bottom of the 2024 semiconductor downturn while running its newest 300mm fabs well below capacity. Now that utilization is rising in Q3 2026, operating margin has already vaulted back to 47.0%.

Cash Flow Forensics: The $9.90B Operating Cash Engine and Stock-Based Compensation Discipline

When we turn from the income statement to the audited cash flow statement, the numbers become even more compelling. Why? Because GAAP net income (7,072 million dollars TTM. 2,400 million dollars in Q3 2026) is currently reduced by 2,263 million dollars of

non-cash depreciation and amortization. Up from just 954 million dollars in fiscal 2021 as RFAB2, LFAB1, and Sherman SM1 entered service. When we add back that non-cash depreciation, trail operating cash flow surged 43.48 percent year-over-year to an all-time record of 9,896 million dollars (including 3,400 million dollars generated in the third quarter of 2026 alone). That represents 139.9 percent cash conversion of GAAP net income.

Notice another forensic detail that separates Texas Instruments from Silicon Valley digital chip designers: stock-based compensation discipline. Over the trailing twelve months ended September 30, 2026, TI's total stock-based compensation expense was just 412 million dollars on 20,766 million dollars of revenue. A tiny 1.98 percent of sales. Compare that to fabless semiconductor peers that routinely hand out 6 to 12 percent of revenue in dilutive stock grants every year. When you subtract 2,578 million dollars of trailing gross capital expenditures from 9,896 million dollars of operating cash flow, standard free cash flow reaches 7,318 million dollars ($7.99 per share), or 8,400 million dollars (~$9.15 per share) when you include the 1,600 million dollars of CHIPS Act cash inflows.

And what did Texas Instruments do with that cash gusher over the past twelve months? It paid 5,174 million dollars in cash dividends. Hiking the quarterly payout by 7 percent in September 2026 to 1.52 dollars per share ($6.08 annualized) for its 23rd consecutive year. Repurchased 636 million dollars of common stock for a total shareholder return of 5,810 million dollars, and still had enough surplus cash in the third quarter of 2026 to repay 500 million dollars of debt net while building its cash balance to 8.10 billion dollars.

> **GLOBAL PARALLEL (USA & ASIA IDM BALANCE SHEETS) • HOW TI FUNDED $19B OF FABS WITHOUT JEOPARDIZING ITS FORTRESS**
>
> Compare how Texas Instruments financed its 2022–2025 fab wave against global peers: US peer Microchip Technology (NASDAQ: MCHP) took on heavy variable-rate acquisition debt that forced dividend stress and compressed ROIC to 6.89%. Meanwhile, Intel (NASDAQ: INTC) suspended its dividend altogether and diluted shareholders via joint-venture financings. Texas Instruments locked in long-dated investment-grade bonds at a weighted-average fixed coupon of just 4.11%, kept $8.10B in cash on hand (5.59x Current Ratio), raised its dividend for the 23rd straight year, and is already retiring debt in Q3 2026.

10-K/10-Q Balance Sheet Teardown: Net Debt, the 100.97% Payout Artifact, and Share Creep

A thorough institutional analyst never glosses over the footnotes. In our forensic audit of Texas Instruments' 2025 Form 10-K, second-quarter 2026 Form 10-Q, June 2, 2026 Form 8-K, and third-quarter 2026 release, cross-checked against ThesisCheck's filing teardown, we examine every single bear-case flag on the balance sheet. First is leverage: at the end of Q2 2026, total carrying debt peaked at 14,052 million dollars before TI repaid 500 million dollars in Q3 2026. That brought principal debt to 13.65 billion dollars (13,555 million standardized carrying value). Against 8,142 million dollars of cash and short-term investments (up from 4,881 million dollars at year-end 2025), net debt is just 5,413 million dollars, or 5.91 dollars per share. That is less than 0.50 times trailing EBITDA (~10.8 billion dollars) and 0.64 times CHIPS-inclusive free cash flow, with a locked-in 4.11 percent average interest coupon that actually sits below the money-market yield TI earns on its 8.1 billion dollar cash pile.

Second is the widely misunderstood '100.97 percent dividend payout ratio' shown on Finviz. Why does that screen show a payout ratio just over 100 percent? Because automated screeners divide the forward or trailing dividend by trailing GAAP net income ($7.72 per share). That is artificially depressed by 2,263 million dollars of non-cash fab depreciation and earlier recovery quarters. Dividends are paid out of cash flow, not GAAP accounting lines. Against 7,318 million dollars of standard trailing free cash flow (or 8,400 million dollars with CHIPS cash), the 5,174 million dollar annual dividend is covered by a healthy 1.41 to 1.62 times. Meaning the 'debt-funded dividend' bear thesis of 2023–2024 is dead and buried.

Third, we note two genuine minor items from the Q2 2026 Form 10-Q and Form 8-K filings that we continue to monitor: first. That is because gross share buybacks were throttled back to 636 million dollars over the trailing twelve months while employees exercised stock options generating 812 million dollars of cash proceeds to the company, average diluted shares crept up by roughly 0.9 percent from 912 million in Q2 2025 to 920 million in Q2 2026 (with Finviz reporting 914.00 million basic shares outstanding on November 6). Second, TI recorded 34 million dollars of first-half 2026 acquisition integration charges

($17 million in Q2) for the Silicon Labs transaction. Both items are minor friction points that will be swamped as 2027 free cash flow scales and buybacks resume in earnest.

> **NARRATOR'S REFLECTION • THE 5.59X CURRENT RATIO FORTRESS**
>
> Pause and admire two liquidity ratios on Texas Instruments' balance sheet as of November 6, 2026: a Current Ratio of 5.59x and a Quick Ratio of 4.05x. Even excluding every dollar of its $4.70B inventory, TI holds more than four times enough liquid cash and receivables to pay off every short-term liability due within the next twelve months.

CHAPTER VII

Deconstructing the $278.59 Price — Reverse DCF, FY2027–2028 Earnings Model & Peer Comps

> *"Price is what you pay. Value is the discounted present stream of owner free cash flow that the asset will hand back to you over the next twenty years."*
>
> — WARREN E. BUFFETT, CHAIRMAN OF BERKSHIRE HATHAWAY (ADAPTED TO THE 300MM CAPITAL CYCLE)

At the November 6, 2026 session close of 278 dollars and 59 cents, the stock market values Texas Instruments' equity at 254.57 billion dollars and its total enterprise value. Including 5.41 billion dollars of net debt. At roughly 259.56 billion dollars. Whenever you and I consider committing hard-earned capital to a quarter-trillion-dollar company, we must ask a blunt, unsparing question: why is the price what it is right now, and what exact future growth rate is already baked into 278 dollars and 59 cents?

As we saw in the Prologue, a naive observer who looks only at trailing twelve-month accounting ratios sees a 36.15 times trailing P/E multiple on 7.72 dollars of diluted EPS and a 34.64 times multiple on 7,318 million dollars of standard trailing free cash flow, and concludes that the stock is expensive for a company whose revenue compounded at 4.10 percent across the 2021–2025 full cycle. But the stock market is a forward discounting mechanism. In the third quarter of 2026 alone, Texas Instruments already achieved an annualized earnings run-rate of 10.48 dollars per share ($2.62 times four) and an annualized standard free cash flow run-rate of 11.75 billion dollars ($2,937 million times four). Meanwhile, guiding the seasonally weak fourth quarter to a midpoint of 2.62 dollars per share. Against consensus fiscal 2027 earnings of roughly 11.65 dollars per share (+28.43 percent forward EPS growth), Texas Instruments trades at 23.90 times forward earnings and a PEG ratio of just 0.68.

To strip away all guesswork, let us walk through our two explicit valuation models from Chapter 7 of the dossier: first, our ten-year Reverse Discounted Cash Flow model solving for the exact free cash flow growth rate implied by 278 dollars and 59 cents. And second, our

bottom-up fiscal 2027 and 2028 earnings and free cash flow model across Bear, Base, and Bull operating regimes.

In Estimate Number One, our ten-year two-stage Reverse DCF model, we equate Texas Instruments' current enterprise value of 259.56 billion dollars (254.15 billion dollars of market equity across 914 million shares at 278.59 dollars plus 5,413 million dollars of net debt) to the present value of ten years of projected free cash flows discounted at a rigorous 9.50 percent Weighted Average Cost of Capital. How do we build that 9.50 percent WACC? We take a 4.45 percent ten-year U.S. Treasury risk-free rate, add a normalized long-term analog beta of 1.10 multiplied by a 4.75 percent equity risk premium, and blend it with TI's 3.57 percent after-tax cost of debt. At Year Ten, we apply a terminal multiple of 22.0 times terminal free cash flow. This corresponds to a 4.55 percent terminal free cash flow yield.

What growth rate does that math reveal? Starting from the trailing twelve-month standard free cash flow base of 7,318 million dollars ($7.99 per share), today's 278 dollar and 59 cent share price embeds an implied ten-year annual free cash flow growth rate of just 8.2 percent per year. Or a mere 6.4 percent per year if you start from the 8,400 million dollar ($9.15 per share) CHIPS-inclusive free cash flow base. Think about how conservative that hurdle is: because third-quarter 2026 standard free cash flow of 2,937 million dollars is already running at 11.75 billion dollars annualized, Texas Instruments only has to sustain its current Q3 2026 cash conversion rate and grow revenue by 5 to 6 percent per year over the next decade to beat the expectations priced into the stock today.

MENTAL MODEL • REVERSE DCF — SOLVING FOR THE HURDLE HEIGHT BEFORE YOU JUMP

"Never argue over whether a stock is 'cheap' or 'expensive' in the abstract. Calculate the exact cash-flow growth rate the current price demands, then ask whether the physical business can clear that bar with room to spare."

Our Reverse DCF sensitivity table shows the exact boundaries for TXN: if standard FCF compounds at only 5.0% annually over the next decade and the terminal multiple compresses to 18.0x, intrinsic value drops to $218.00/share. At 8.2% annual FCF growth and 22.0x terminal FCF, intrinsic value equals today's $278.59. And if TXN compounds standard FCF at 11.5% annually. Simply growing revenue 10% into existing 300mm cleanrooms while keeping CapEx at $2.5B–$3.5B. With a 24.0x terminal multiple, present intrinsic value reaches $342.00/share.

Our Bottom-Up FY2027–FY2028 Earnings Model and $325.00 Base Target

Now let us walk through Estimate Number Two: our bottom-up operating model for fiscal 2027 and fiscal 2028. With Q3 2026 actual revenue at 6.10 billion dollars and Q4 2026 guided to a midpoint of 6.05 billion dollars, Texas Instruments is on pace to finish full-year fiscal 2026 with roughly 21.76 billion dollars in revenue. For fiscal 2027, our Base Case projects revenue of 25,350 million dollars. That represents 16.5 percent year-over-year growth over full-year 2026, but notice this crucial fact: 25.35 billion dollars is only 3.9 percent above the 24.40 billion dollar annualized run-rate that Texas Instruments already achieved in the third quarter of 2026! In other words, modest sequential expansion across Industrial (+18 percent YoY), Data Center (+35 percent YoY), and Automotive (+12 percent YoY), reinforced by a 150-basis-point net pricing tailwind from 2027 contract renewals, easily delivers our 25.35 billion dollar revenue forecast.

What happens to margins and earnings at 25.35 billion dollars of fiscal 2027 revenue? Applying Julie Knecht's 75 to 85 percent incremental revenue fall-through ex-depreciation as wafer loadings rise at RFAB2, LFAB1, and Sherman SM1, fiscal 2027 gross margin expands 150 basis points from Q3 2026 levels to 65.5 percent (16,604 million dollars of gross profit). Subtracting 4,150 million dollars of operating expenses (16.4 percent of sales) yields 12,454 million dollars of operating income, or a 49.1 percent operating margin. After subtracting 180 million dollars of net interest and applying a 13.5 percent effective tax rate

(right inside Knecht's 13 to 14 percent guidance range), GAAP net income reaches 10,617 million dollars, or 11.65 dollars per diluted share across 911 million average shares, with standard free cash flow reaching roughly 10.50 dollars per share.

Applying a historical mid-cycle valuation multiple of 28.0 times fiscal 2027 consensus earnings of 11.65 dollars per share (and 31.0 times standard free cash flow) yields our twelve-month Base Case target price of 325 dollars per share (+16.7 percent capital appreciation from 278.59 dollars, or +18.8 percent total return including the 6.08 dollar dividend). In our Bull Case (30 percent probability), stronger AI Data Center and Industrial restocking lifts fiscal 2027 revenue to 26.50 billion dollars, gross margin to 66.8 percent, and EPS to 12.30 dollars (with fiscal 2028 EPS sightlines above 14.00 dollars). This at 30.0 times FY27E earnings produces a 368 dollar target (+32.1 percent). In our Bear Case (20 percent probability), an industrial order pause holds FY27 revenue to 23.00 billion dollars, compresses gross margin to 61.5 percent, and pulls the multiple to 20.0 times $11.25 EPS, establishing a 225 dollar downside floor (-19.2 percent).

Quantitative Peer Comparison: Why TXN Commands a ROIC and Quality Premium Over ADI, NXPI, MCHP & ON

Let us cross-check our valuation against the live November 6, 2026 peer comparison matrix across Texas Instruments and its four primary U.S.-listed analog and mixed-signal peers: Analog Devices (NASDAQ: ADI), NXP Semiconductors (NASDAQ: NXPI), Microchip Technology (NASDAQ: MCHP), and ON Semiconductor (NASDAQ: ON). Listen closely to these numbers. That is because they demolish the myth that Texas Instruments is overpriced relative to its closest large-cap rival.

Compare Texas Instruments ($278.59, $254.57B market cap) directly against its closest mega-cap analog peer, Analog Devices ($365.71, $177.21B market cap). On trailing twelve-month P/E, Texas Instruments trades at 36.15 times. More than seven turns cheaper than Analog Devices at 43.40 times! On enterprise-value-to-EBITDA, Texas Instruments trades at 23.97 times. Cheaper than Analog Devices at 26.34 times. On price-to-free-cash-flow, Texas Instruments trades at 34.64 times. Cheaper than Analog Devices at 35.72 times. And on

dividend yield, Texas Instruments pays 2.08 to 2.18 percent ($6.08 annualized), nearly double Analog Devices' 1.17 percent yield.

Yet while you pay a lower trailing P/E, lower EV/EBITDA, and lower P/FCF for Texas Instruments than for Analog Devices, look at the superior business quality you get in return: Texas Instruments generates a 41.12 percent trailing operating margin (and 47.0 percent in Q3 2026) versus ADI's 35.89 percent. A 22.04 percent return on invested capital versus ADI's 10.25 percent. And a 39.50 percent return on equity versus ADI's 12.23 percent. Meanwhile, against cyclical peers NXP Semiconductors (19.48x TTM P/E, 29.92% operating margin, 14.63% ROIC), Microchip Technology (87.88x GAAP TTM P/E, 16.23%–21.57% operating margin, 6.89% ROIC, stock down 44.50% from high), and ON Semiconductor (36.99x TTM P/E, 38.13% gross margin, 6.02% ROIC, stock down 52.89% from high), Texas Instruments stands alone as the fortress compounder of the sector.

GLOBAL PARALLEL (USA & EUROPE VALUATION COMPS) • WHY CAPTIVE 300MM ROIC (22.04%) BEATS GOODWILL-HEAVY ROLL-UPS

Why is Analog Devices' ROIC (10.25%) less than half of Texas Instruments' ROIC (22.04%) despite similar 60%+ gross margins? Because ADI paid massive acquisition premiums to buy Linear Technology ($14.8B) and Maxim Integrated ($21B), bloating its invested capital base with goodwill while outsourcing wafers to TSMC. TI built its 300mm fabs organically (or bought Micron's Lehi fab and Qimonda's tools at distressed prices) and has not done a major deal since National Semiconductor in 2011. Allowing every dollar of 300mm operating leverage to compound directly on a clean tangible capital base.

CHAPTER VIII

Weinstein Stage Analysis, Moving-Average Confluence & Institutional Ownership Structure

> *"Never buy a stock just because its fundamentals look wonderful if the tape is pulling back beneath its 30-week moving average. Let the moving averages and volume tell you when institutional buyers have built the floor."*
>
> — STAN WEINSTEIN'S STAGE ANALYSIS DISCIPLINE, ADAPTED BY THE INSTITUTIONAL RESEARCH DESK

We have now established beyond a shadow of a doubt that Texas Instruments is a world-class analog fortress whose fundamental engine, revenue, gross margin, operating margin, and free cash flow, inflected sharply higher on October 21, 2026. If you and I were pure academic theorists who ignored market structure and price action, we would simply close our eyes and hit the market buy button at 278 dollars and 59 cents. But on an institutional desk, we never divorce fundamental research from the geometry of the tape. Great fundamentals tell you what to own. Stan Weinstein Stage Analysis and moving-average confluence tell you when and where to pull the trigger.

When we inspect Texas Instruments' daily and weekly charts on November 6, 2026, we encounter a textbook analytical puzzle that every serious investor must know how to solve. On our automated morning run of stock_snapshot.py, the mechanical scanner flagged Texas Instruments as STAGE_4 / AVOID_STAGE4_OR_BROKEN. Why did a computer script apply a Stage 4 tag to a stock that just reported +28 percent year-over-year revenue growth and +251.8 percent trailing free cash flow growth? Because over the five trading sessions ended November 6, 2026, a broad semiconductor sector sell-off pulled Texas Instruments down 9.44 percent to close at 278 dollars and 59 cents. Putting the stock 1.79 percent below its 30-week simple moving average of 283 dollars and 68 cents, 1.25 percent below its 10-week exponential moving average of 282 dollars and 13 cents, and a mere 0.18 percent (49 cents) below its 50-day simple moving average of 279 dollars and 8 cents.

Under a rigid mechanical filter, any stock that closes simultaneously beneath its 30-week and 50-day simple moving averages triggers an automatic warning. Our job as human analysts is

never to ignore that warning, nor to follow it blindly without context, but to look under the hood of the multi-timeframe chart and diagnose whether Texas Instruments is suffering a true Stage 4 structural breakdown. Like peers Microchip Technology (sitting 26.16 percent below its declining 200-day moving average) and ON Semiconductor (sitting 23.01 percent below its declining 200-day moving average). Or whether TXN is simply in an intermediate Stage 3 pullback testing primary support within a powerful long-term uptrend.

Four pieces of hard quantitative evidence prove that Texas Instruments is in a healthy Stage 3 digest-and-retest pullback rather than a Stage 4 breakdown. First, over the trailing twelve months, TXN has rallied 71.18 percent (+59.92 percent year-to-date), climbing from a 52-week low of 152 dollars and 73 cents to a 52-week high of 334 dollars and 3 cents, with a three-year gain of 88.09 percent and a ten-year gain of 310.41 percent. Second, the primary long-term trendlines. The 200-day simple moving average at 262 dollars and 28 cents, the 40-week exponential moving average at 262 dollars and 28 cents, and the 200-day exponential moving average at 261 dollars and 86 cents. Are all sloping steadily upward and sit 6.22 percent below the current share price. Third, on November 6, the stock held right on top of its monthly Classic S1 support pivot at 277 dollars and 54 cents while trading volume contracted to 6.20 million shares. Just 0.83 times its 30-day average volume of 7.43 million shares and 0.63 times its 10-day average of 9.83 million shares. And fourth, long-only institutions own an overwhelming 92.21 percent of the float (with net positive institutional accumulation of +0.08 percent). Meanwhile, short interest is a negligible 2.22 percent of float (20.22 million shares, or 3.08 days to cover).

MENTAL MODEL • DISTINGUISHING A STAGE 3 PULLBACK FROM A STAGE 4 BREAKDOWN

"A stock sitting 6.2% above a rising 200-day moving average on drying volume (0.83x) is catching its breath. A stock sitting 25% below a falling 200-day moving average on heavy distribution is in the hospital."

Compare TXN against its own semiconductor group on November 6, 2026: Analog Devices (-2.32% vs 200D SMA), Microchip (-26.16% vs 200D SMA), and ON Semiconductor (-23.01% vs 200D SMA) have all broken below their 200-day moving averages. Texas Instruments (+6.22% above its rising $262.28 200-day SMA) is the relative-strength general of the analog industry resting on its 50-day line ($279.08).

Why We Wait for Support: Semiconductor Breadth (17% > 50D EMA) and the Overhead Moving-Average Stack

If Texas Instruments' primary 200-day uptrend is intact and volume on the pullback is light, why does our institutional discipline command a verdict of Wait for Support rather than placing an immediate market buy order at 278 dollars and 59 cents tonight? Two concrete technical facts dictate our patience. First, look at TheWrap's live US Stage 2 industry breadth ledger for Semiconductors: while 86 percent of the broader semiconductor universe has outperformed the S&P 500 over the multi-year cycle, short-term group breadth has washed out to just 26.0 percent of semiconductor stocks trading above their 200-day exponential moving average and a mere 17.0 percent trading above their 50-day exponential moving average.

When four out of five stocks in a sector are trading below their 50-day moving averages, institutional sector ETF redemptions and quantitative basket selling create a downward undertow that frequently drags even the strongest fundamental leader slightly below its 50-day moving average to test its rising 200-day moving average before the sector bottoms. Second. That is because Texas Instruments dropped 9.44 percent over the past week, its short-term moving averages are currently stacked overhead as ceiling resistance: the 10-day simple moving average sits at 301 dollars and 34 cents, the 10-day exponential moving average at 294 dollars and 66 cents, the 20-day exponential moving average at 292 dollars and 35 cents, the 20-day simple moving average at 291 dollars and 10 cents, the 50-day exponential moving average at 285 dollars and 33 cents, the 30-week simple moving average at 283 dollars and 68 cents, and the 10-week exponential moving average at 282 dollars and 13 cents. Above those moving averages sit the monthly Classic Middle Pivot at 295 dollars and 31 cents, the Classic R1 pivot at 318 dollars and 60 cents, and the 52-week high at 334 dollars and 3 cents.

Now map the two-tier structural floor directly beneath our feet at 278 dollars and 59 cents. Tier-One tactical support is right where the stock closed on November 6: the 50-day simple moving average at 279 dollars and 8 cents and the monthly Classic S1 support pivot at 277 dollars and 54 cents, nestled inside our scanner's local support band of 273 dollars and 38 cents to 281 dollars and 70 cents. If semiconductor group weakness nudges TXN through

277 dollars and 54 cents, Tier-Two primary structural support awaits just 5.85 percent lower at the triple confluence of the rising 200-day simple moving average ($262.28), the rising 40-week exponential moving average ($262.28), and the rising 200-day exponential moving average ($261.86).

Check the volatility math on that distance: Texas Instruments' 14-day Average True Range (ATR) is 12 dollars and 24 cents, or 4.39 percent of the share price (with weekly volatility at 5.32 percent, monthly volatility at 3.81 percent, daily RSI at 43.19, and weekly RSI at 51.98). A move from 278 dollars and 59 cents down to the 262 dollar and 28 cent 200-day moving average is only 1.33 times a single day's ATR! By waiting for either a weekly close reclaiming the 30-week SMA at 283 dollars and 70 cents or a support bounce inside 262 dollars and 28 cents to 279 dollars and 9 cents. And anchoring our hard invalidation stop at 255 dollars and 80 cents (0.5x ATR14, or $6.12, below the rising 200-day SMA). We eliminate the risk of getting whipsawed by a routine sector shakeout.

> **GLOBAL PARALLEL (PHILADELPHIA SOX & EUROPEAN SEMI TAPES) • HOW INSTITUTIONAL DESKS TRADE A 17% BREADTH WASHOUT**
>
> Across both the US Philadelphia Semiconductor Index (SOX) and European semiconductor leaders like ASML (NASDAQ: ASML), Infineon (ETR: IFX), and STMicroelectronics (EPA: STM), history shows that when sector breadth above the 50-day EMA drops below 20%, the washout is in its final innings. But the final shakeout wick often tags the 200-day moving average of the resilient leaders. Waiting for either the 200-day bounce ($262.28–$268.00) or the 30-week reclaim ($283.70) lets the market prove the low is in.

CHAPTER IX

Steelman Bull vs. Bear Debate, Probability-Weighted Scenarios & No-FOMO Execution Plan

> *"Invert, always invert: before you ask how much money you will make if your thesis is right, ask what could break the machine, where your thesis becomes invalid, and how much portfolio heat you will lose if you are wrong."*
>
> — CHARLIE MUNGER'S INVERSION PRINCIPLE APPLIED TO INSTITUTIONAL RISK MANAGEMENT

We have reached the final chapter of our masterclass journey through Texas Instruments Incorporated. Also, this is where analysis transforms into decision. Before you and I ever commit capital to a stock, our institutional rulebook demands that we step across the aisle, put on the hat of a ruthless short-seller or skeptical red-team risk officer, and build the strongest honest Bear Case we can construct against our own thesis. If our Bull Case cannot survive an unsparing collision with the Bear Case across every operational and financial dimension, we have no business owning the stock.

Let us conduct that six-dimension Steelman Bull versus Bear debate right now. Dimension One is the durability of the Q3 2026 breakout in Industrial (+40 percent year-over-year) and Data Center (+95 percent year-over-year). The Red-Team Bear argues that part of third-quarter 2026's 28 percent year-over-year revenue surge reflects temporary post-destocking inventory replenishment and urgent 'golden screw' kitting orders as lead times ticked higher, warning that if global manufacturing PMIs or hyperscaler AI capital spending slow in late 2027, order rates will mean-revert. The Bull counters, and our independent judgment agrees, that fourth-quarter 2026 guidance of 5.80 to 6.30 billion dollars is five to seven percentage points stronger than normal winter seasonality. This proves that four years of rising analog content per factory machine and 10,000 part numbers across 48-volt AI server racks represent genuine structural expansion.

Dimension Two is 300-millimeter wafer fab operating leverage versus fixed-cost depreciation risk. The Bear points directly to Texas Instruments' Form 10-K and Q2 2026 Form 10-Q footnotes, noting that owning captive fabs saddles TI with 2,180 million dollars of fixed

annual fab depreciation (2,263 million dollars total D&A) plus factory start-up costs. This compressed gross margins to 57.02 percent in fiscal 2025 when loadings were low. The Bull replies that fixed-cost leverage cuts both ways: now that cleanroom shells at RFAB2, LFAB1, and Sherman SM1 are built and wafer loadings are rising, that exact fixed-cost structure delivered nearly 90 percent incremental gross margin in Q3 2026 (lifting gross margin 280 basis points sequentially to 64.0 percent and operating margin to 47.0 percent), with five-year GAAP tool depreciation scheduled to roll off long before the thirty-year physical life of the machines.

> **GLOBAL PARALLEL (INSTITUTIONAL CAPITAL ALLOCATORS) • BUFFETT'S BURLINGTON NORTHERN & INFINEON'S DRESDEN PAYBACK CURVE**
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> *"The market punishes a capital-intensive business while the tracks are being laid, and then re-rates it as an inevitable compounding royalty the year the capital spending stops."*
>
> We have seen this exact movie across global markets: whether in US rail and energy infrastructure, Germany's Infineon (ETR: IFX) ramping its 300mm Villach power fab, or Texas Instruments after its 2009–2011 RFAB1 and National Semiconductor investments. Between 2012 and 2021, after TI finished its previous investment wave, free cash flow per share compounded at 12% annually and TXN shares rose more than sixfold. Q3 2026 marks the opening curtain of that exact harvest phase on a 300mm footprint four times larger.

Dimension Three is free cash flow inflection versus the 13.65 billion dollar debt load and dividend payout ratio. The Bear cites Finviz's 100.97 percent trailing GAAP dividend payout ratio and the 13.65 billion dollars of gross debt accumulated during the 2022–2025 construction wave. Our forensic audit shows why the Bull wins this point decisively: trailing twelve-month CapEx has already fallen to 2,578 million dollars ($463 million in Q3) while operating cash flow hit an all-time record of 9,896 million dollars, propelling standard free cash flow up 251.8 percent to 7,318 million dollars ($8.40 billion with CHIPS Act cash). The 5.17 billion dollar dividend is now 1.41 to 1.62 times covered by actual cash flow, net debt is just 5.41 billion dollars at a 4.11 percent fixed coupon, and TI already repaid 500 million dollars of debt in Q3.

Dimension Four pits 2027 annual contract pricing power against Chinese domestic analog substitution. Bears worry that Chinese design houses like Silergy and SG Micro will

commoditize standard 200-millimeter analog sockets. Yet as Haviv Ilan confirmed on October 21, TI's China revenue grew faster than the company average in Q3 2026 thanks to high-spec Chinese EV powertrains. Meanwhile, virtually zero foundational analog wafer capacity was built outside China and Taiwan over the past four years. Allowing TI to embed positive pricing into 2027 supply-guarantee contracts. Dimension Five acknowledges a minor Bear point from the Q2 2026 Form 10-Q/8-K: diluted shares crept up ~0.9 percent from 912 million to between 914 million and 920 million as option exercises ($812M TTM) exceeded throttled buybacks (-$636M TTM) alongside $34 million of H1 Silicon Labs charges. A trend that 2027 excess cash flow will reverse. And Dimension Six synthesizes valuation and technicals: 23.90x forward P/E (0.68x PEG) is compelling, but sitting 1.79 percent below the 30-week SMA ($283.68) demands our Wait for Support posture.

Probability-Weighted 12-Month Scenario Math and Position-Sizing Mechanics

Let us combine our fundamental scenarios from Chapter 7 with our technical architecture from Chapter 8 into a probability-weighted expected return calculation from the November 6, 2026 close of 278 dollars and 59 cents. We assign a 30 percent probability to our Bull Case of 368 dollars per share (based on $26.50B of FY27 revenue, 66.8 percent gross margin, and $12.30 EPS at 30.0x, or +34.28 percent total return with the 2.18 percent dividend yield), contributing 110 dollars and 40 cents to expected value.

We assign a 50 percent probability to our Base Case of 325 dollars per share (based on $25.35B of FY27 revenue, 65.5 percent gross margin, $11.65 EPS at 28.0x, and $10.50 FCF per share, or +18.84 percent total return with dividends), contributing 162 dollars and 50 cents. And we assign a 20 percent probability to our Red-Team Bear Case of 225 dollars per share (based on $23.00B of FY27 revenue, 61.5 percent gross margin, and $11.25 EPS at 20.0x, or -19.24 percent downside), contributing 45 dollars.

Summing those three probability-weighted contributions ($110.40 + $162.50 + $45.00) yields a twelve-month expected share price of 317 dollars and 90 cents (+14.11 percent price appreciation and +16.29 percent expected total return including the $6.08 annual dividend). Against our hard invalidation stop at 255 dollars and 80 cents (-8.18 percent risk

from $278.59), our 325 dollar Base Target offers a 2.04-to-1 reward-to-risk ratio at the current print, rising to 3.85-to-1 if accumulated near the rising 200-day moving average ($262.28 to $268.00).

Under our scoring configuration, TXN sits in the Core Alpha strategy bucket at Conviction Tier Two (5.0 to 10.0 percent weight band. 75 basis points portfolio risk budget). Dividing our 0.75 percent risk budget by the 8.18 percent stop distance from 278.59 dollars to 255.80 dollars produces a raw risk-unit weight of 9.17 percent. We set our target allocation at 7.0 percent of net worth, and once trending in Stage 2, allow it to compound up to our 22.0 percent runaway-winner ceiling.

The Three Exact Decision Triggers Governing Our Capital

Let us close our conversation by stating the three exact price and event triggers that govern how you and I will deploy capital in Texas Instruments.

Bullish Entry Trigger Number One is our Weekly Trend Reclaim (Immediate Buy): watch for NASDAQ: TXN to record a Friday weekly close at or above 283 dollars and 70 cents on volume above 7.43 million shares. That reclaims both the 30-week simple moving average ($283.68) and the 10-week exponential moving average ($282.13). Buy the first 3.5 percent tranche on that Friday close, and add the second 3.5 percent tranche (to reach the full 7.0 percent weight) above the 20-day moving average at 291 dollars and 10 cents, with our hard stop at 255 dollars and 80 cents.

Bullish Entry Trigger Number Two is our 200-Day Moving Average Support Bounce (Accumulate in Zone): if sector weakness pushes TXN into the rising 200-day simple moving average ($262.28) and 40-week exponential moving average ($262.28) zone between 262 dollars and 28 cents and 268 dollars, watch for a daily reversal close back above 270 dollars with daily RSI above 35.0. Accumulate the full 7.0 percent Tier-2 stake inside 262 dollars and 28 cents to 272 dollars, capturing a 3.85-to-1 reward-to-risk ratio toward our 325 dollar base target against the 255 dollar and 80 cent stop.

Bearish Invalidation Trigger Number Three is our Breakdown or Guide Miss (Exit / Stand Aside): if TXN closes below 255 dollars and 80 cents on heavy volume, or if Q1 2027

revenue guidance in January 2027 falls below 5.60 billion dollars with gross margin under 61.0 percent, exit or cancel all orders immediately and wait for a new Stage 1 base near the 225 dollar bear floor. With 300-millimeter cleanrooms standing in Richardson, Lehi, and Sherman, Industrial and Data Center setting records, and free cash flow surging past eight billion dollars, you and I have the complete blueprint in hand. Now we let the tape come to our price.

THE EXECUTION PLAYBOOK • SUMMARY CHECKLIST FOR YOUR TRADING DESK

- Trigger #1 (Buy Strength): Friday close >= $283.70 on volume > 7.43M shares -> Buy 3.5% tranche, add 3.5% above $291.10 (total 7.0% weight).
- Trigger #2 (Buy Support Value): Pullback test of rising 200D SMA ($262.28–$268.00) + daily reversal close > $270.00 -> Buy 7.0% weight (3.85:1 R:R to $325.00).
- Trigger #3 (Hard Invalidation): Close Exit / Stand Aside.
- Long-Term Compounding Rule: Hold 7.0% CORE_ALPHA stake up to 22.0% runaway-winner cap. Never trim solely for RSI > 70 while price holds above the rising 21-week EMA.

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This Masterclass Companion Book was authored for private study, audiobook narration, and disciplined portfolio execution.

Verified against the US Research Ledger • Session Cutoff 2026-11-06